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The Effect of Gravity Darkening during Exoplanet Transits

RESEARCH THESIS

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Abstract

The three-dimensional spin–orbit architecture of exoplanetary systems provides a powerful probe of the formation and migration pathways of close-in giant planets. While spectroscopic techniques such as the Rossiter–McLaughlin effect and Doppler tomography have successfully constrained projected spin–orbit misalignments, their effectiveness is severely limited for hot, rapidly rotating stars due to extreme rotational line broadening. In this regime, gravity-darkened transit photometry offers a unique and complementary method for constraining the full three-dimensional system geometry.

In this work, we investigate whether gravity-darkened transit modelling of space-based photometry can robustly constrain the true spin–orbit angle, ψ, for the extreme hot-Jupiter system HAT-P-70. We analyse high-precision photometry from TESS Sector 32 and identify statistically significant asymmetric features in the transit light curve that cannot be reproduced by conventional limb-darkened transit models. These asymmetries motivate the inclusion of stellar oblateness and gravity darkening in the transit modelling.

A physically self-consistent framework describing stellar rotation, oblateness, and rotation-dependent gravity darkening is developed and implemented within the STARRY modelling package, which enables the computation of transit light curves for non-uniform, latitude-dependent stellar brightness distributions. Using forward modelling and Markov Chain Monte Carlo techniques, we explore the sensitivity of the observed transit morphology to key stellar and orbital parameters, including the stellar inclination, projected spin–orbit angle, and gravity-darkening properties. Two physically motivated stellar configurations—one adopted from the literature and one derived independently from rotational constraints—are explored to assess the robustness of the inferred system geometry.

We find that gravity-darkened transit modelling provides meaningful constraints on the stellar inclination and three-dimensional spin–orbit architecture of HAT-P-70, despite the presence of significant parameter degeneracies. The results consistently favour a strongly misaligned, retrograde orbit, in agreement with previous spectroscopic studies and supporting high-eccentricity migration scenarios for the formation of hot Jupiters around hot stars. This work demonstrates both the power and the limitations of gravity-darkened photometry and highlights its importance as a complementary tool for probing the three-dimensional architectures of exoplanetary systems orbiting rapidly rotating stars.

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1. Introduction

The architecture of planetary systems reflects the physical processes that govern planet formation, migration, and long-term dynamical evolution (*Lin et al., 1996; Chatterjee et al., 2008; Nagasawa et al., 2008; Batygin, 2012; Rice et al., 2022*). In particular, the three-dimensional orientation between a planet's orbital plane and its host star's rotational axis — the spin–orbit angle — has emerged as a powerful diagnostic for distinguishing between competing migration pathways of close-in giant planets (*Winn et al., 2005; Triaud et al., 2010; Albrecht et al., 2012*).

Spectroscopic techniques such as the Rossiter–McLaughlin effect and Doppler tomography have provided important constraints on projected spin–orbit misalignments (*Rossiter, 1924; McLaughlin, 1924; Gaudi & Winn, 2007; Collier Cameron et al., 2010; Johnson et al., 2017*). However, radial-velocity precision their effectiveness is significantly reduced for hot, rapidly rotating stars, where severe rotational line broadening limits and complicates line-profile analysis (*Winn et al., 2010; Gray, 2021; Zhou et al., 2019*). In this observational regime, photometric techniques exploiting gravity-darkened transit signatures offer a powerful and, in some cases, unique means of recovering the full three-dimensional architecture of exoplanetary systems (*von Zeipel, 1924; Barnes, 2009; Barnes et al., 2011; Masuda, 2015; Ahlers et al., 2020*).

This project investigates the extent to which gravity-darkened transit modelling of space-based photometry can robustly constrain the true spin–orbit geometry of such systems, with particular focus on the extreme hot-Jupiter system HAT-P-70 (*Zhou et al., 2019; Cauley & Ahlers, 2022*).

1.1 Scientific Context

Understanding the physical processes that shape the brightness distribution of stars is fundamental to the study of transiting exoplanets. Several mechanisms can contribute to surface–brightness inhomogeneities, including limb darkening (*Claret, 2017*), magnetic activity such as starspots and faculae (*Basri & Nguyen, 2018*), granulation driven by convection (*Beeck et al., 2013*), and stellar pulsations. While each of these effects can influence the observed transit signal to some extent, gravity darkening is exceptional in producing a global, latitude-dependent brightness gradient across the stellar disk. This phenomenon becomes especially important in hot, rapidly rotating stars, making gravity darkening a powerful diagnostic tool for spin–orbit geometry in exoplanetary systems (*von Zeipel 1924; Barnes 2009; Espinosa Lara & Rieutord, 2012*).

The gravity darkening effect occurs when rapid stellar rotation deforms a star into an oblate spheroid, with an equatorial radius larger than the polar radius. The resulting variation in effective surface gravity causes the poles—with higher gravity—to become hotter and brighter, while the equator—with lower gravity—becomes cooler and dimmer. This relationship is described by the classical von Zeipel law, which predicts a temperature dependence of $T_{\text{eff}} \propto g^{\beta}_{\text{eff}}$ with $\beta = 0.25$ for radiative envelopes (*von Zeipel, 1924*). More recent models developed by Espinosa Lara & Rieutord (2011) incorporate the degree of stellar flattening—the parameter f—and provide physically consistent gravity-darkening exponents that vary with rotation rate, while preserving the fundamental pole-to-equator brightness contrast.

These gravity-darkened intensity variations introduce characteristic asymmetries into the transit light curves of planets orbiting such stars. As a planet moves across regions of differing surface brightness, the light curve departs from the symmetric shape expected for a spherical, limb-darkened star. These asymmetries can provide valuable, though not complete, constraints on the stellar spin inclination and the orbital geometry of the system (*Barnes, 2009; Ahlers et al., 2020*). In particular, they help access the three-dimensional spin–orbit angle (ψ), a quantity often inaccessible through spectroscopic techniques alone, especially for rapidly rotating stars with broadened spectral lines.

The magnitude of gravity darkening depends strongly on stellar rotation and internal structure, making it most prominent in hot A- and early F-type stars. These stars lie above the Kraft break, a transition near $T_{eff} \approx 6250$ K where magnetic braking becomes inefficient due to the thinning or absence of convective envelopes (*Kraft, 1967; Dawson, 2014*). With weak magnetic braking, these stars retain high rotation rates throughout their lifetimes, become significantly oblate, and therefore exhibit pronounced gravity-darkened brightness distributions. In contrast, cooler stars below the Kraft break spin down rapidly and typically rotate too slowly for gravity darkening to produce an observable transit signature, hence not flattening the star enough to detect significant temperature variations across the surface.
For this reason, studies of gravity-darkened transits—and this project in particular—focus on hot A/F stars, where the effect is both physically significant and observationally detectable. Hot Jupiter systems around such stars offer especially valuable laboratories. A well-studied example is KELT-9b, where the extreme rotation of the B9.5/A0-type host star produces a strongly asymmetric, gravity-darkened transit as shown in **Figure 1** (*Ahlers et al., 2020*).

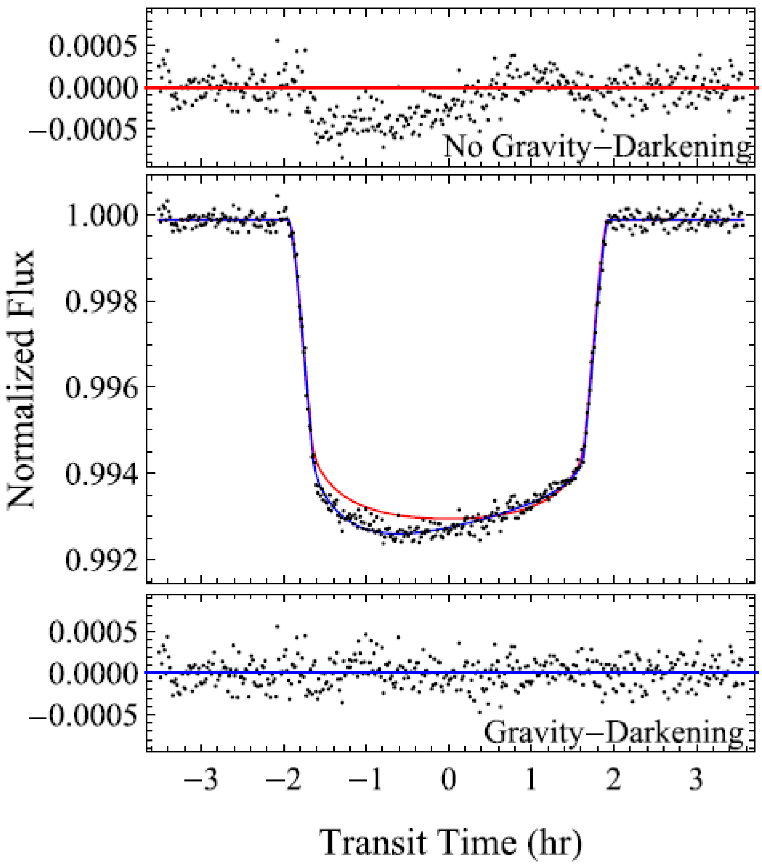

Figure 1. Phase-folded primary transit of the hot Jupiter KELT-9b. The red curve shows the expected transit assuming only standard limb darkening, while the blue curve shows the best-fitting gravity-darkened model. The pronounced asymmetry in the observed transit indicates that the planet begins its transit near the brighter stellar pole and moves toward the cooler equatorial region. (Figure adapted from *Ahlers et al., 2020.*)

Systems like this help demonstrate how gravity-darkening analysis can reveal information about stellar obliquity, orbital alignment, and the dynamical histories of close-in giant planets. These concepts provide the necessary foundation for the modelling and analysis undertaken in this project.

Against the theoretical framework developed above, the HAT-P-70 system represents an exceptionally favourable target for investigating gravity-darkened transits and spin–orbit geometry. The host star HAT-P-70 is a rapidly rotating late A-type star with a published effective temperature of approximately $T_{\mathrm{eff}} \approx 8450$ K, a stellar radius of $R_{\star} \approx 1.86\ R_{\odot}$, and a projected rotational velocity of nearly 100 km s^{-1} (*Cauley & Ahlers, 2022*). These properties place the star well above the Kraft break and firmly within the regime where gravity darkening is expected to be both physically significant and observationally detectable. Cauley & Ahlers (2022) further report that the star exhibits substantial rotational deformation, with an inferred oblateness of $f \approx 0.08$, and a pronounced pole-to-equator temperature contrast of almost 850 K, corresponding to a polar temperature of approximately 8670 K and an equatorial temperature near 7830 K (*Cauley & Ahlers, 2022*).

The planetary companion HAT-P-70b is an inflated hot Jupiter with a published radius of $R_{\mathrm{p}} \approx 1.87\ R_{\mathrm{j}}$ on a short-period 2.74-day orbit (*Zhou et al., 2019*). Its large size produces deep, high signal-to-noise transits, while the short orbital period allows repeated events within the TESS observing baseline. Both papers also report a strongly misaligned and retrograde orbit, with a projected spin–orbit angle of $\lambda \approx 113^{\circ}$, indicating that the planet crosses a wide range of stellar latitudes (*Cauley & Ahlers, 2022; Zhou et al., 2019*) and because of this geometry, it maximises the observable imprint of gravity darkening on the transit light curve.

Importantly, these published stellar and planetary parameters provide the initial physical framework for the present study. In this project, the values reported by Cauley & Ahlers (2022) for the stellar properties and by Zhou et al. (2019) for the planetary and orbital parameters are adopted as starting inputs for the transit and gravity-darkening models. Through detailed modelling of the TESS light curves, these parameters will be systematically tested and refined, allowing an independent assessment of their consistency with the observations and an evaluation of whether the gravity-darkening signature recovered in this work agrees with, or deviates from, previously published results.

Finally, the same extreme stellar rotation that enhances the gravity-darkening signal also severely limits the precision of traditional spectroscopic methods. The broad and blended spectral lines of HAT-P-70 significantly complicate Rossiter–McLaughlin and Doppler-tomographic measurements (*Zhou et al., 2019*). Consequently, HAT-P-70 lies precisely in the observational regime where gravity-darkened transit modelling becomes not only complementary but essential to help the stellar inclination $i_{\star}$ and the full three-dimensional spin–orbit angle ψ.

Taken together, the extreme stellar rotation, strong gravity-darkening signature, large planetary radius, short orbital period, and severe spin–orbit misalignment make HAT-P-70 an outstanding natural laboratory for probing the physics and dynamics of close-in giant planets, and an ideal focal system for the analysis presented in this project.

1.2 Research Question and Aim of the Project

Although gravity-darkened transit modelling has been shown to help provide access to the three-dimensional spin–orbit geometry of exoplanetary systems (*Barnes, 2009; Ahlers et al.,* *2020; Cauley & Ahlers, 2022*), the reliability of the derived parameters depends on both the quality of the photometric data and the modelling methods adopted. For systems hosting extremely rapid rotators such as HAT-P-70, conventional spectroscopic techniques—in particular the Rossiter–McLaughlin effect and Doppler tomography—become increasingly difficult to apply due to severe rotational line broadening and the associated loss of radial velocity precision (*Zhou et al., 2019*). As a result, gravity-darkened photometric modelling becomes not only complementary but essential for recovering the remaining geometric information.

Motivated by the interesting observational properties of HAT-P-70 outlined in **Chapter 1.1**, this project aims to evaluate the extent to which gravity-darkened transit modelling of space-based photometry can independently and robustly constrain the full three-dimensional spin–orbit architecture of this system when combined with existing spectroscopic constraints.

The primary research question guiding this investigation is therefore:

"*Can gravity-darkened transit modelling of TESS data for HAT-P-70 robustly constrain the true three-dimensional spin–orbit angle, ψ, when combined with spectroscopic constraints***?"**

To address this question, the project is structured around the following objectives:

1. **Detection of gravity-darkening signatures**
   To identify and characterise the gravity-darkening imprint in the TESS transit light curves of HAT-P-70 by isolating asymmetric features that cannot be reproduced by standard limb-darkened models.
2. **Transit modelling using STARRY**
   To model the observed transits using the STARRY framework, incorporating stellar oblateness, gravity darkening, and the projected orbital geometry in order to reproduce the observed light-curve morphology.
3. **Derivation of system geometry and computation of ψ**
   To derive the orbital inclination i, the stellar spin inclination $i_\star$, and the projected spin–orbit angle λ from the best-fitting models, and to compute the true three-dimensional spin–orbit angle ψ using the formalism of Masuda (2015),
4. **Physical interpretation in the context of planetary migration**
   To interpret the recovered spin–orbit configuration within the broader theoretical framework of hot Jupiter migration and dynamical evolution, and to assess which migration pathways are most consistent with the observed system architecture.

Through this approach, the project seeks both to provide an independent characterisation of the HAT-P-70 system and to assess the broader applicability and limitations of gravity-darkened transit modelling as a tool for probing the three-dimensional architectures of exoplanetary systems orbiting rapidly rotating stars.

2 Literature Review and Theory

This chapter summarises the theoretical and observational framework of gravity-darkened transit modelling and spin–orbit alignment studies. The aim is not to reproduce the full literature review presented in the earlier stages of this project, but rather to establish the physical principles, key results, and measurement techniques necessary for the analysis of HAT-P-70.

2.1 Spin–Orbit Alignment and Migration Theories

The spin–orbit angle ψ describes the three-dimensional orientation between a planet's orbital axis and the rotational axis of its host star (**Figure 2**).

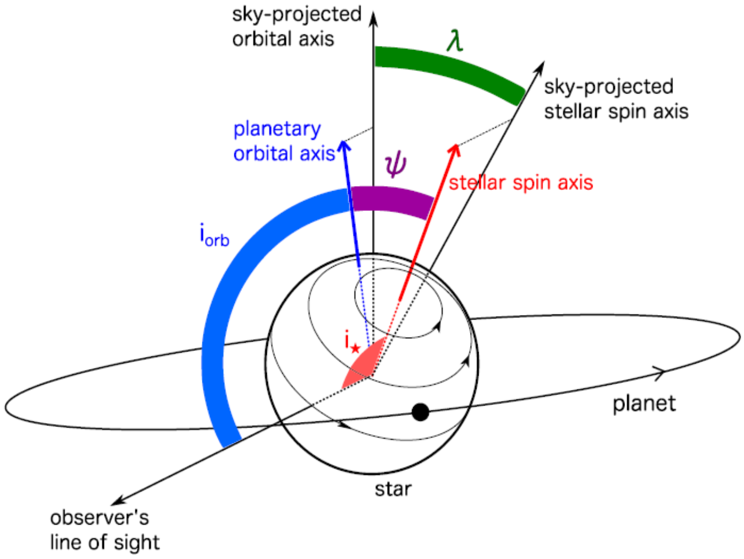

Figure 2. Illustrates the geometric relationships between the relevant angles. The orbital inclination, i_{orb} (blue), is defined as the angle between the planet's orbital axis and the observer's line of sight. The stellar inclination, $i_{\star}$, is the angle between the observer's line of sight and the true stellar rotation axis. The angle between the planetary orbital axis and the stellar spin axis is the three-dimensional spin–orbit angle ψ(purple). When these angles are projected onto the plane of the sky, they yield the sky-projected obliquity λ(green). (Adapted from *Masuda, 2015*.)

This quantity is a fundamental diagnostic of planetary system evolution. Classical formation theory predicts near alignment ($\psi \approx 0°$) as planets form within a co-rotating protoplanetary disk (*Winn et al., 2005*). However, observations reveal a wide diversity of spin–orbit configurations, including highly misaligned and retrograde orbits, particularly among hot Jupiters (*Triaud et al., 2010; Albrecht et al., 2012*).

Several mechanisms have been proposed to explain these misalignments. Disk-driven migration may preserve alignment unless the disk itself becomes tilted by external torques, such as from a stellar companion (*Lin et al., 1996; Batygin, 2012*). In contrast, dynamical interactions, including planet–planet scattering (*Chatterjee et al., 2008; Nagasawa et al., 2008*), Kozai–Lidov oscillations induced by distant companions, and secular chaos can excite large orbital eccentricities and inclinations. When combined with tidal dissipation at small orbital separations, these processes can drive planets onto short-period orbits that are both circularised and strongly misaligned (*Fabrycky & Tremaine, 2007; Wu & Lithwick, 2011*). Over longer time-scales, tidal dissipation within the star may act to damp spin–orbit misalignments, particularly in close-in systems (*Winn et al., 2010*).

A strong observational trend exists with stellar temperature. Systems orbiting stars hotter than the Kraft break exhibit a much broader distribution of misalignment angles compared with cooler stars (*Winn et al., 2010; Albrecht et al., 2012; Rice et al., 2022*). This behaviour is commonly attributed to differences in stellar internal structure: cool stars possess substantial convective envelopes that enable efficient tidal dissipation and realignment, while hot stars above the Kraft break have radiative envelopes and weak magnetic braking, resulting in much longer tidal realignment timescales.

Recent work by Rice et al. (2022) has refined this picture by demonstrating that the obliquity–temperature relationship depends critically on the orbital eccentricity of the planetary companion. Using a population-level analysis of hot Jupiters with measured sky-projected obliquities, Rice et al. (2022) showed that the sharp transition in obliquity at the Kraft break is present only for planets on circular orbits. In contrast, hot Jupiters on eccentric orbits display a wide range of misalignments across the full stellar temperature range, with no statistically significant discontinuity at the Kraft break. This result is illustrated in **Figure 3.**

Within the framework of high-eccentricity migration, this distinction arises naturally. Dynamical interactions initially place planets on highly eccentric and often highly inclined orbits, producing a population of systems with large spin–orbit misalignments independent of stellar temperature. Tidal dissipation subsequently acts to reduce orbital eccentricity and, on longer time-scales, to damp stellar obliquity. Because tidal dissipation is far more efficient in cool stars than in hot stars, circularised planets around cool stars are expected to appear preferentially aligned at the present epoch, even if they were initially misaligned. Eccentric systems, by contrast, represent an earlier or less-processed evolutionary stage in which tidal dissipation has not yet had time to significantly alter either eccentricity or obliquity (*Rice et al., 2022*).

This interpretation provides a unified explanation for the observed spin–orbit distribution of hot Jupiters, without requiring multiple distinct formation pathways for aligned and misaligned systems. Instead, a single dominant migration channel—high-eccentricity migration—combined with differential tidal damping shaped by stellar structure can reproduce the observed population-level trends. Systems orbiting hot stars above the Kraft break are therefore expected to retain much of their primordial spin–orbit architecture, making them particularly valuable probes of planetary migration physics.

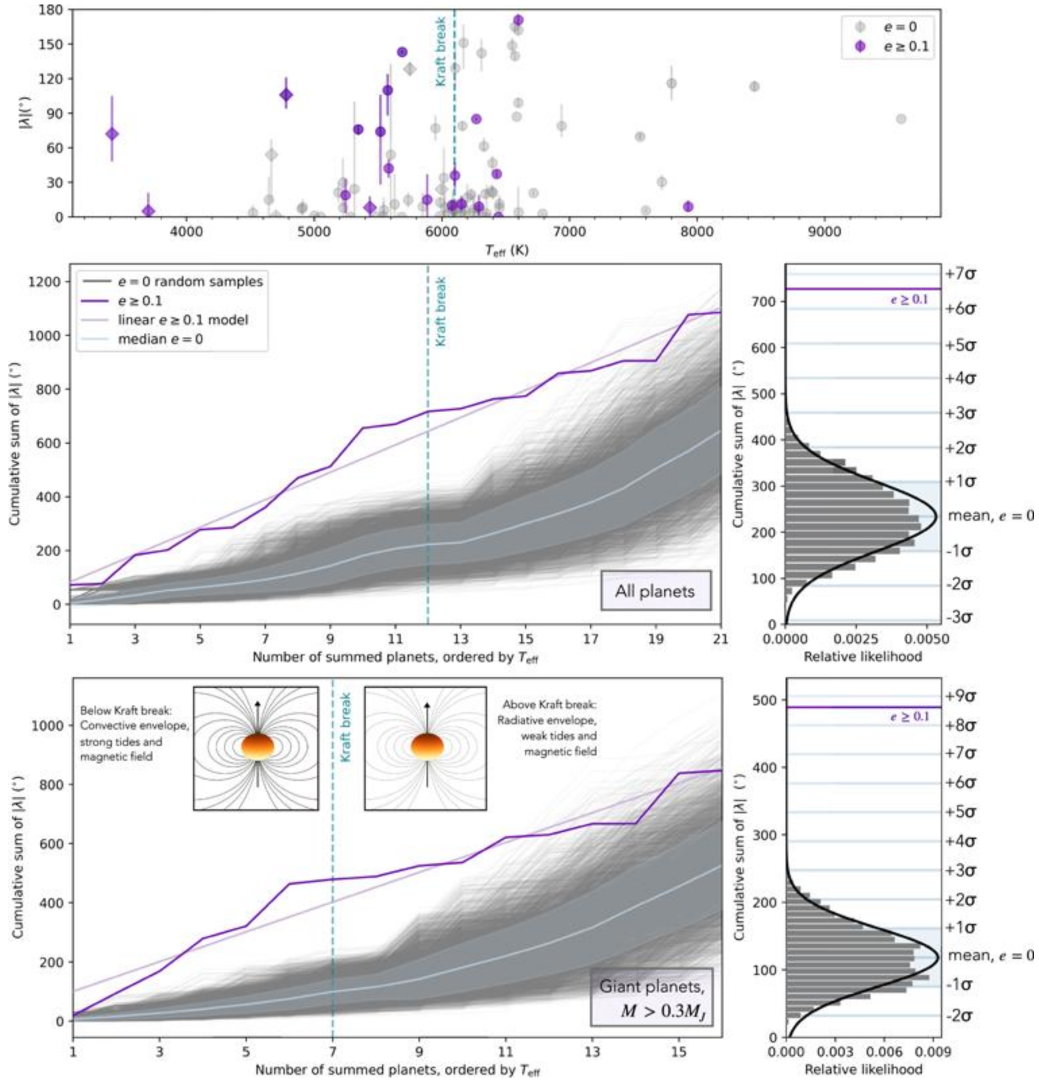

Figure 3. Distribution of sky-projected stellar obliquities (|λ|) as a function of host star effective temperature for hot-Jupiter systems, separated by orbital eccentricity. The **upper panel** displays the full sample of measured systems, with eccentric planets (e ≥ 0.1) shown in purple and circular planets (e $= 0$) in grey; the vertical dashed line denotes the Kraft break at $T_{\rm eff} \approx 6100$ K. The **middle and lower panels** show the cumulative sums of |λ|, ordered by increasing stellar temperature, for all planets and for giant planets ($M > 0.3\ M_J$), respectively. Circular systems exhibit a pronounced increase in spin–orbit misalignment above the Kraft break, consistent with inefficient tidal realignment in hot stars, whereas eccentric systems display no comparable transition. This distinction provides strong evidence that the observed obliquity distribution of hot-Jupiter systems is primarily shaped by high-eccentricity migration followed by differential tidal damping. (Adopted from *Rice et al., 2022*)

2.2 Transit Geometry and Observables

The morphology of a transit light curve is governed by the orbital inclination i_{orb}, the stellar inclination $i_{\star}$, the sky-projected spin–orbit angle λ, and the impact parameter b (*Seager & Mallén-Ornelas, 2003; Fabrycky & Winn, 2009*). For rapidly rotating, oblate stars, gravity darkening introduces a strong latitude-dependent surface brightness distribution, such that the transit chord intersects regions of non-uniform intensity. This produces characteristic light-curve asymmetries whose shape and amplitude depend sensitively on the system's three-dimensional geometry (*Barnes, 2009; Masuda, 2015*).

The impact parameter plays a central role: transits probing higher stellar latitudes (near the rotational poles) are typically deeper and shorter, while equatorial transits are shallower and longer due to the reduced surface brightness and increased chord length across the oblate stellar disk (**Figure 4**) (*Barnes, 2009*). The orientation of the transit chord relative to the stellar rotation axis, parameterised by λ, determines which latitudes are sampled during transit, while the stellar inclination $i_{\star}$ sets the apparent orientation of the gravity-darkened brightness pattern on the sky. Stellar oblateness further modifies the transit geometry by altering both the projected stellar shape and the local surface gravity, amplifying asymmetries in misaligned systems. Accurate modelling of limb darkening, stellar oblateness, and gravity darkening is therefore essential for extracting reliable geometric constraints from transit light curves (Howarth, 2011; Espinoza & Jordán, 2015).

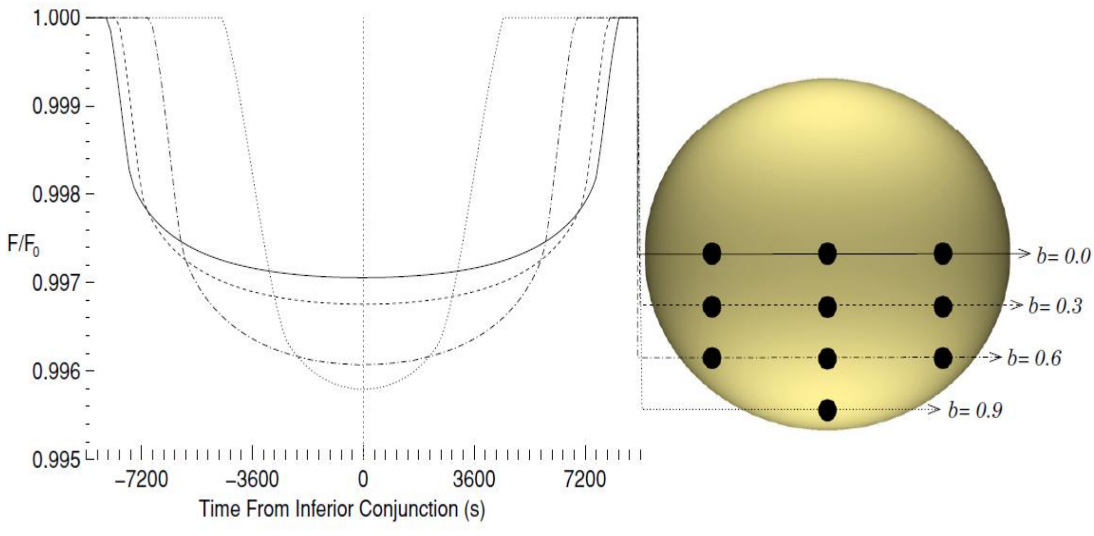

Figure 4. Shows synthetic transit light curves for a Jupiter-sized planet (1 R_J) orbiting an oblate star at 0.05 AU in a spin–orbit aligned configuration. The four curves correspond to different transit impact parameters. Transits occurring closer to the stellar poles are noticeably deeper but shorter in duration, reflecting the higher photospheric temperature at the poles caused by the gravity-darkening effect**.** (Adapted from *Barnes, 2009*)

2.3 Measurement Techniques

While the preceding sections establish the physical importance of spin–orbit alignment and the observable consequences of transit geometry—particularly in gravity-darkened stars—the extraction of these parameters from real systems requires specialised observational techniques. The asymmetries and distortions predicted by theory are subtle and cannot be fully constrained through photometry alone, especially for rapidly rotating host stars. Consequently, a combination of spectroscopic and photometric methods has been developed to measure the relative orientations of stellar rotation and planetary orbits.

2.3.1 Rossiter–McLaughlin Effect

The Rossiter–McLaughlin (RM) effect arises when a transiting planet blocks different regions of a rotating stellar disk, producing a temporary distortion in the observed stellar radial velocity (RV) curve (*Rossiter, 1924; McLaughlin, 1924; Gaudi & Winn, 2007*). As the planet successively obscures the blueshifted and redshifted hemispheres of the star, the centroid of the rotationally broadened stellar absorption lines is temporarily shifted, producing an anomalous RV signal superimposed on the Keplerian orbital motion. An example of this effect is shown in **Figure 5**.

The RM signal depends sensitively on the geometry of the transit chord across the stellar disk. Its amplitude and morphology are governed by the projected stellar rotation velocity $v\sin i_\star$, the transit impact parameter b, the adopted limb-darkening prescription, and the sky-projected spin–orbit angle λ (*Ohta et al., 2005; Hirano et al., 2011*). By modelling the shape and sign of the RV anomaly, the RM effect provides robust constraints on λ, enabling direct observational tests of planetary migration theories. RM measurements have revealed a striking diversity of spin–orbit architectures, including strongly misaligned and retrograde hot-Jupiter systems (*Winn et al., 2010; Triaud et al., 2010*).

Despite its success, the RM effect is intrinsically a disk-integrated measurement, compressing spatial information across the stellar surface into a single RV time series. As a result, RM analyses suffer from degeneracies—most notably between λ and $v \sin i_\star$—and are sensitive to assumptions about limb darkening and stellar line broadening. These limitations become increasingly severe for hot, rapidly rotating stars, where extreme rotational broadening reduces RV precision and weakens the observable RM signal. This motivates the use of a more spatially resolved spectroscopic technique, such as Doppler tomography, explained in **Chapter 2.3.2**

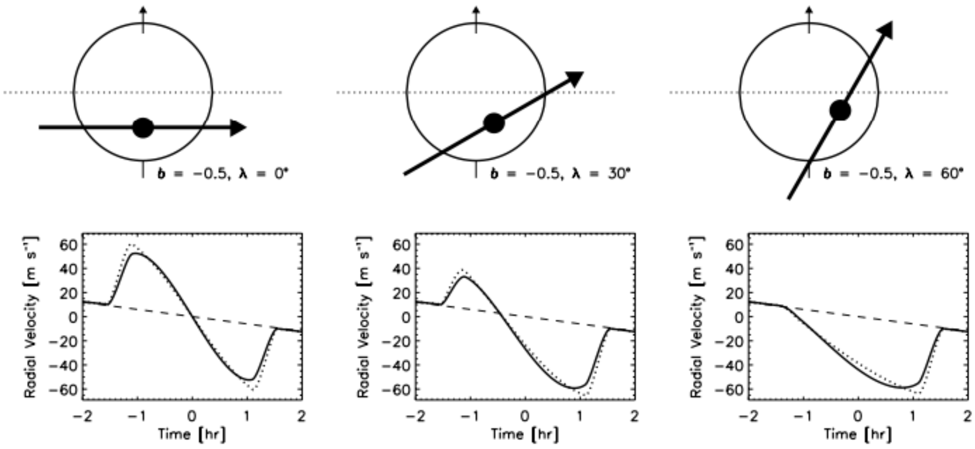

Figure 5. Example of how different projected orbital paths of a transiting exoplanet across a rotating star produce distinct signatures in the observed radial velocity curve. The star is shown rotating anticlockwise, with the hemisphere approaching the observer appearing blue-shifted and the receding hemisphere red-shifted. Although the transit impact parameter is identical in all cases, each panel corresponds to a different value of the sky-projected spin–orbit angle, λ, illustrating how variations in orbital alignment led to characteristic distortions in the radial velocity signal. (Adapted from *Gaudi & Winn*, *2007*).

2.3.2 Doppler Tomography

Doppler tomography extends the principles underlying the Rossiter–McLaughlin effect by directly resolving distortions in stellar spectral line profiles during planetary transit. Rather than measuring only the net shift of the line, Doppler tomography tracks the time-dependent deformation of the rotationally broadened stellar line profile caused by the planet blocking light from specific velocity regions on the stellar surface (*Collier Cameron et al., 2010*). A case study for the hot Jupiter HAT-P-41b is represented in **Figure 6**.

As the planet transits the stellar disk, it produces a moving deficit in the line profile—commonly referred to as a Doppler shadow—that traces the projected path of the planet across the stellar surface in velocity space. By modelling the trajectory, width, and contrast of this shadow, Doppler tomography enables precise measurements of the sky-projected spin–orbit angle λ and provides direct constraints on the projected stellar rotation velocity $v\sin i_\star$ (*Johnson et al., 2015; Johnson et al., 2017*).

This technique is particularly powerful for rapidly rotating and early-type host stars, for which traditional RV-based RM measurements are compromised by broad spectral lines and reduced RV precision. In such systems, tomographic analyses often yield tighter and more robust constraints on λ than the RM effect alone and reduce degeneracies between geometric and stellar parameters (*Gaudi et al., 2017*).

While it provides detailed information about the projected geometry of the system, it does not independently constrain the stellar inclination $i_\star$. Consequently, even when combined with RM measurements, Doppler tomography alone cannot fully recover the true three-dimensional spin–orbit angle ψ, motivating the complementary use of gravity-darkened transit photometry explained in **Chapter 2.3.3**.

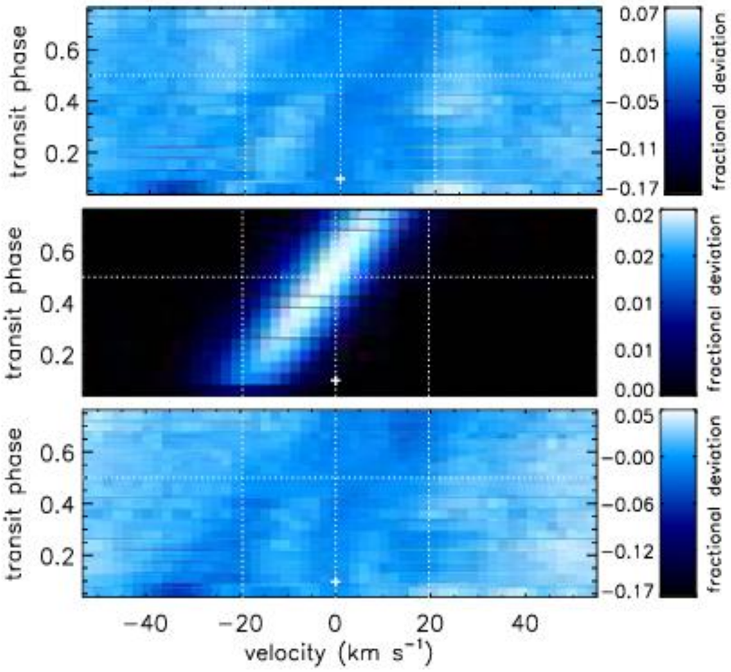

Figure 6 shows a Doppler tomography measurement of HAT-P-41b using time-series line profile residuals from five partial transits observed with the Hobby–Eberly Telescope. The bright streak in the upper panel indicates a prograde orbit, the middle panel shows the best-fitting model, and the lower panel displays the residuals after model subtraction, with remaining structure likely due to systematic effects. (Adapted from

2.3.3 Gravity-Darkened Photometry

Ultimately, gravity-darkened transit modelling directly constrains the stellar inclination $i_\star$ by fitting asymmetric light curves produced by non-uniform stellar brightness distributions (*Barnes, 2009; Masuda, 2015; Ahlers et al., 2020*). When combined with RM and Doppler tomography, this approach enables full three-dimensional reconstruction of the system geometry.

2.4 The ψ Equation and Motivation

The ultimate goal of spin–orbit alignment studies is the determination of the system's true three-dimensional obliquity, ψ, which quantifies the angle between the stellar rotation axis and the planetary orbital angular momentum vector. As discussed in **Chapters 2.1–2.3,** theoretical models of planetary migration predict large distributions of ψ, while observable transit signatures and spectroscopic distortions provide indirect access to the underlying geometry. These elements are unified by the following relation (*Masuda, 2015*):

$$\cos\,\psi = \cos\,i_\star \cos\,i + \sin\,i_\star \sin\,i \cos\,\lambda. \quad \textbf{Equation} \quad \textbf{1}$$

Here, the orbital inclination i is determined from transit photometry, the sky-projected spin–orbit angle λ is constrained via the Rossiter–McLaughlin effect or Doppler tomography, and the stellar inclination $i_{\star}$ is uniquely accessible through gravity-darkened transit modelling. Only by combining these complementary techniques can the full three-dimensional architecture of the system be recovered.

3 Data and Methodology

This chapter presents the observational data, analysis pipeline, and physical modelling framework developed to investigate gravity darkening in the HAT-P-70 system. **Chapter 3.1** describes the selection of HAT-P-70 as a target, the retrieval of TESS photometry, and the full preprocessing procedure, including data cleaning, detrending, phase folding, and correction for third-light contamination. **Chapter 3.2** establishes a statistical baseline by fitting conventional limb-darkened transit models to the processed light curve, systematically evaluating the quality of the fits and identifying persistent asymmetries that cannot be reproduced by standard symmetric models. **Chapter 3.3** develops a self-consistent physical framework for stellar rotation, oblateness, and gravity darkening. Using a custom Python implementation, the stellar inclination, rotational deformation, gravity-darkening coefficient, and surface temperature distribution of HAT-P-70 are derived over the full range of physically allowed configurations. **Chapter 3.4** applies these physically motivated stellar models within the STARRY framework to generate gravity-darkened transit simulations. The sensitivity of the observable transit morphology to key stellar and orbital parameters is explored, and the resulting synthetic light curves are directly compared with the TESS observations. Finally, **Chapter 3.5** presents the final stage of the analysis, in which the physically motivated STARRY models are coupled with forward modelling and Markov Chain Monte Carlo (MCMC) techniques to infer the posterior distributions of the system parameters. This stage provides the quantitative constraints on the stellar inclination, spin–orbit geometry, and gravity-darkening properties of HAT-P-70 and forms the basis for the scientific conclusions of this work

3.1 TESS Data and Preprocessing

3.1.1 Target Selection and System Overview

The HAT-P-70 system was selected as the primary target for this investigation due to its exceptional suitability for studying gravity darkening in transiting exoplanet systems. HAT-P-70 is an early-type A-star rotating at a projected velocity of nearly 100 km s^{-1}, placing it well above the Kraft break and firmly within the regime where gravity darkening is expected to be strong. In combination with this rapid stellar rotation, the system hosts a large hot Jupiter (HAT-P-70b) on a short-period orbit of 2.744 days, producing frequent and deep transits that are readily observable with space-based photometry (*Cauley & Ahlers, 2022*).

The principal stellar and planetary parameters adopted for this stage of light curve analysis are taken from the NASA Exoplanet Archive and the detailed spectroscopic and photometric analysis of Cauley & Ahlers (2022) and from Zhou et al. (2019), and were reported in **Table 1**. These include a stellar effective temperature of approximately 8450 K, a stellar spehrical radius of 1.86 $R_{\odot}$, a planetary radius of 1.87 R_J, and a large projected spin–orbit misalignment of $\lambda \approx$ 113°, making HAT-P-70 a very promising known system for detecting gravity-darkened transit signatures.

Name of System	HAT-P-70	NASA Exoplanet Archive
Name of Planet	HAT-P-70b	NASA Exoplanet Archive
Planet Radius (R_J)	$1.87^{+0.15}_{-0.10}$	*Zhou et al. (2019)*
Planet Mass (M_J)	$< 6.78\ (3\sigma)$	*Zhou et al. (2019)*
Orbital Period (days)	2.744	*Zhou et al. (2019)*
T_{14} (days)	$0.1450^{+0.0028}_{-0.0020}$	*Zhou et al. (2019)*
Lambda (λ)	$113.1^{+5.1}_{-3.4}$	*Zhou et al. (2019)*
Orbital Inclination (i_{orb})	$96.50^{+1.42}_{-0.91}$	*Zhou et al. (2019)*
Impact Parameter (b)	$-0.629^{+0.081}_{-0.054}$	*Zhou et al. (2019)*
$R_p/R_\star$	$0.09887^{+0.00133}_{-0.00095}$	*Zhou et al. (2019)*
$a/R_\star$	$5.45^{+0.29}_{-0.49}$	*Zhou et al. (2019)*
v sin i (km/s)	99.9 ± 0.6	*Cauley & Ahlers (2022)*
Star Mass ($M_\odot$)	$1.890^{+0.013}_{-0.010}$	*Cauley & Ahlers (2022)*
Star Radius ($R_\odot$)	$1.858^{+0.119}_{-0.091}$	*Cauley & Ahlers (2022)*
Star T_{eff} (K)	8450^{+540}_{-690}	*Cauley & Ahlers (2022)*
Gravity (log g)	$4.181^{+0.055}_{-0.063}$	*Cauley & Ahlers (2022)*
TESS Magnitude	9.298 ± 0.019	*Zhou et al. (2019)*
Metallicity [Fe/H]	0.0	*Cauley & Ahlers (2022)*
Right Ascension	04h58m12.56s	NASA Exoplanet Archive
Declination	+09d59m52.73s	NASA Exoplanet Archive

Table 1. Stellar and planetary parameters compiled from the literature, primarily from Cauley & Ahlers (2022) and Zhou et al. (2019), together with photometric properties retrieved from the NASA Exoplanet Archive. These parameters are adopted as input for the initial modelling and analysis of the TESS light curve using the *Lightkurve* Python package.

3.1.2 TESS Observation and Data Availability

Photometric observations of HAT-P-70 were obtained from the Transiting Exoplanet Survey Satellite (*TESS; Ricker et al., 2015*). A systematic inspection of all available TESS sectors revealed that Sector 32, observed in 2020 with a 120-second cadence, provided the only dataset of sufficient quality and continuity for the present analysis.

SearchResult containing 1 data products.

#	mission	year	author	exptime s	target_name	distance arcsec
0	TESS Sector 32	2020	SPOC	120	399870368	0.0

Figure 7. Snapshot produced as part of this analysis using the *Lightkurve* Python package, illustrating the availability of TESS photometric data for HAT-P-70. Only a single observing sector, Sector 32 (2020), is currently accessible

New observations of HAT-P-70 were scheduled for release in early January 2026 as part of TESS Sector 98, in which the system was again included as a science target (*MIT TESS, 2025*). According to the MIT-TESS mission archive, Sector 98 comprises observations obtained during orbits 205 through 208, covering the period from 9 November 2025 to 5 January 2026, and represents the second of two sectors spanning four spacecraft orbits. However, at the time of completing this work, the Sector 98 data products had not yet been publicly released. Given the advanced stage of the analysis and the extensive modelling already undertaken, it was not feasible to delay the project to incorporate these new observations. Consequently, this study proceeds using only the Sector 32 dataset, which remains fully adequate for achieving the scientific objectives of this work.

3.1.3 Light Curve Processing and Contamination Correction

The TESS photometry was processed using the *Lightkurve* Python package. Initial cleaning involved removing measurements flagged by the TESS pipeline as poor quality, followed by careful manual inspection to identify additional outliers arising from cosmic rays, scattered light, and spacecraft systematics. The resultant data displayed clear and stable transit signals across all observing segments, with only a small number of poor-quality data points. Individual transits were observed approximately every three days, consistent with the expected orbital period of HAT-P-70b. Any anomalous points that deviated significantly from the general light-curve behaviour were identified and masked during this stage to ensure that they did not influence the final analysis. The full preprocessing and cleaning procedure is illustrated in **Figure 8**.

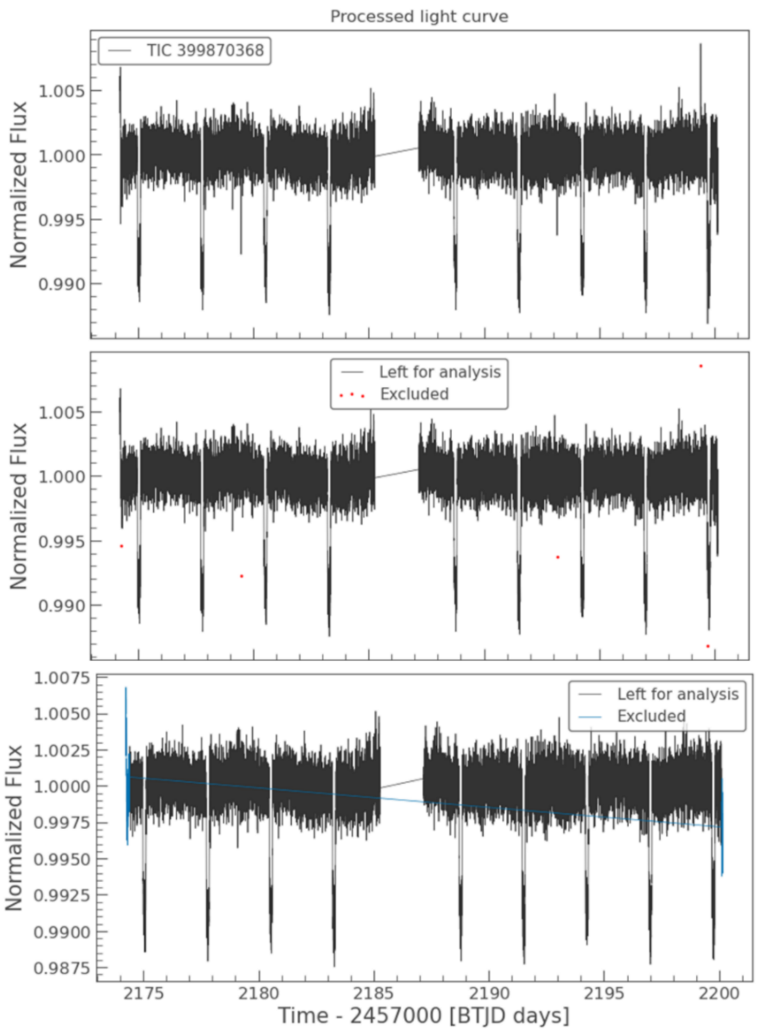

Figure 8. Overview of the full data acquisition and cleaning procedure for HAT-P-70. The **top panel** shows all transits from TESS Sector 32 using the raw, unprocessed photometric data. The **middle panel** displays the same dataset after excluding anomalous data points, which are highlighted in red. The **bottom panel** presents the adopted occultation masks, indicating regions where anomalous points were identified and removed because they could introduce deviations from the true transit light curve.

Long-term trends were removed by fitting a smooth baseline function to the out-of-transit data, thereby isolating the transit signal while preserving its intrinsic morphology. The cleaned light curve was then phase-folded using the published orbital period, producing the final transit profile in **Figure 9**. The resulting phase-folded transit exhibits a subtle but measurable asymmetry, deviating from the symmetric shape typically expected for systems without gravity darkening. This feature provided the indication that gravity darkening may play a significant role in shaping the observed transit signal of HAT-P-70b.

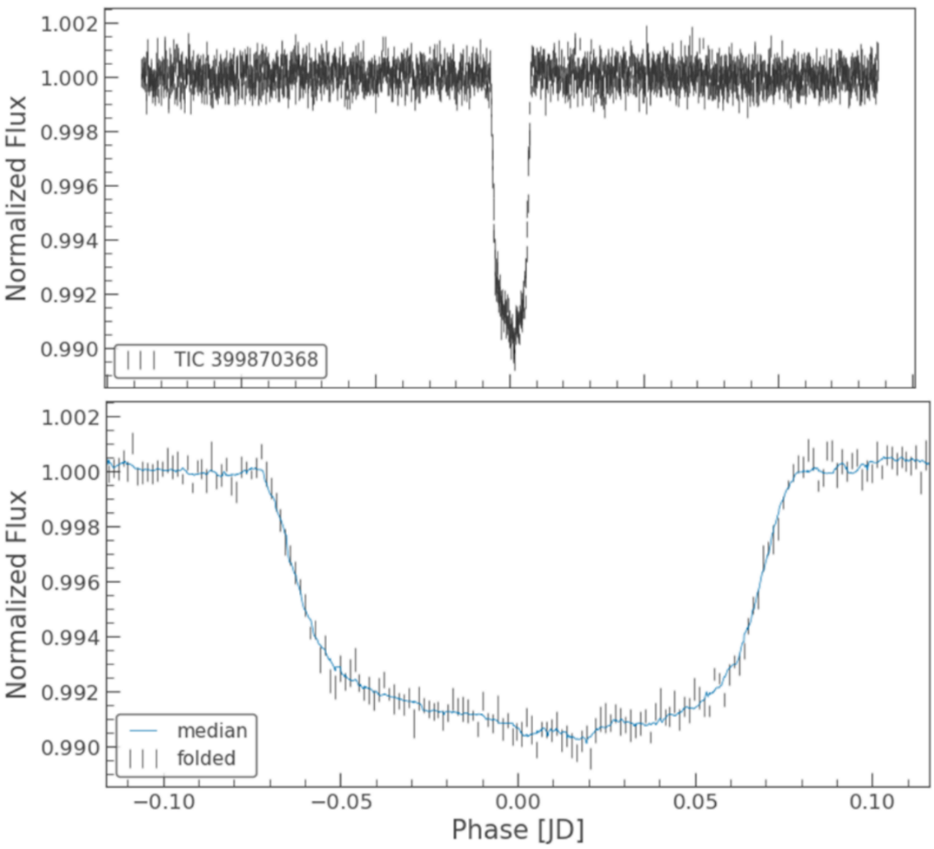

Figure 9. Phase-folded TESS light curve of HAT-P-70b showing an asymmetric transit profile. The **top panel** presents the full phase-folded light curve with associated photometric uncertainties. The **bottom panel** shows a zoomed-in view of the same transit, included to better visualise the asymmetric signal, which is less apparent in the full-scale view. The transit exhibits a deeper ingress and a comparatively shallower egress, consistent with expectations for a gravity-darkened stellar surface.

An important complication in the analysis arises from the presence of a nearby contaminating star. Zhou et al. (2019) demonstrated that a neighbouring star approximately 0.75 magnitudes fainter in the TESS band contributes a non-negligible fraction of the measured flux for HAT-P-70, thereby artificially reducing the observed transit depth. Although the TESS pipeline partially mitigates this effect through its choice of photometric aperture and systematic corrections, careful examination of the Target Pixel File diagnostics remains essential (*Jenkins et al., 2016*).

The TESS Target Pixel File headers provide two key parameters used to assess flux contamination: CROWDSAP, which quantifies the fraction of total flux in the aperture originating from the target star, and FLFRCSAP, which gives the fraction of the target's intrinsic flux that falls within the selected aperture (*Jenkins et al., 2016*). For HAT-P-70 in Sector 32, the CROWDSAP value indicates that approximately 13% of the flux within the aperture arises from contaminating sources, with the remaining 87% originating from the target star.

Further guidance on applying this correction was provided through detailed discussions with my supervisor, which helped clarify the practical implementation of the renormalisation procedure. Because the extracted light curve had initially been normalised to the combined flux of both the target and contaminating stars, a flux level of unity corresponded to $0.13F_{\mathrm{contam}} + 0.87F_{\star}$. To recover the intrinsic stellar flux of HAT-P-70, the following transformation was applied to the phase-folded light curve, where the *y-axis* represents the normalised flux and σ denotes the associated uncertainties, which were also renormalised accordingly.

$$y_{\mathrm{corr}} = \frac{y-0.13}{0.87}, \sigma_{\mathrm{corr}} = \frac{\sigma}{0.87}. \qquad \textbf{Equation 2}$$

This procedure subtracts the contaminating contribution and rescales the remaining flux so that unity corresponds to the intrinsic brightness of HAT-P-70 alone. The uncertainties were scaled by the same factor to preserve the correct relative error structure. After applying this correction, the renormalised light curve was saved and adopted as the input for all subsequent modelling and Markov Chain Monte Carlo analyses presented in this work in **Chapter 4**

3.2 Light Curve Fitting and Statistical Assessment

Before investigating more complex physical interpretations of the observed transit asymmetry, it is essential to establish how well the data can be explained by a conventional transit model. This section, therefore, focuses on assessing the ability of a standard limb-darkened transit model to reproduce the observed TESS light curve of the HAT-P-70 system.

The primary aim of this analysis is to provide a statistically robust baseline against which more physically detailed models may be compared. By systematically fitting a standard transit model and evaluating the resulting residuals, any persistent discrepancies between the model and the data can be identified and characterised. Such discrepancies, when statistically significant, may indicate the presence of additional astrophysical effects influencing the transit morphology.

Using the processed and phase-folded photometry described in **Chapter 3.1**, a sequence of increasingly flexible fits was performed. The quality of each fit was evaluated using chi-squared statistics, allowing the limitations of the conventional limb-darkened model to be quantified prior to introducing gravity darkening in subsequent sections.

3.2.1 Fitting Framework and Data Preparation

The light-curve fitting was carried out using a Python-based modelling notebook provided by the project supervisor. This framework implements an analytic transit model for a limb-darkened, spherically symmetric star, in which the transit morphology is determined by the planet-to-star radius ratio $R_p/R_\star$, the scaled semi-major axis $a/R_\star$, the orbital inclination i_{orb} (or impact parameter b), the mid-transit time T_0 and a set of limb-darkening coefficients. The framework also provides tools for parameter optimisation and statistical evaluation, allowing both fixed-parameter baseline models and progressively more complex fits to be constructed and directly compared with the observational data.

The input light curve used for all fits was the final processed TESS photometry obtained following the detrending, normalisation, and phase-folding steps outlined in **Chapter 3.1**. This ensured that the fitting procedure operated on a dataset optimised for astrophysical interpretation rather than instrumental systematics.

An initial synthetic transit model was generated using published stellar and planetary parameters for the HAT-P-70 system compiled from the NASA Exoplanet Archive, together with values reported by Zhou et al. (2019) and Cauley & Ahlers (2022), as summarised in **Table 1**. This baseline model represents the expected symmetric transit morphology in the absence of gravity darkening and serves as the reference case for subsequent analysis.

Stellar limb darkening was modelled using a quadratic limb-darkening law. The corresponding coefficients were adopted from the stellar atmosphere calculations of Claret (2017), which provide limb- and gravity-darkening coefficients for the TESS photometric bandpass over a wide range of effective temperatures, surface gravities, metallicities, and microturbulent velocities. The coefficients were retrieved via the VizieR (https://vizier.unistra.fr/) catalogue service.

The limb-darkening coefficients were selected using the literature values of the stellar effective temperature, surface gravity, and metallicity of HAT-P-70. The adopted coefficients were $a_{LD} = 0.1870$ and $b_{LD} = 0.2672$. These parameters were held fixed throughout the fitting procedure. Fixing the limb-darkening coefficients ensures that variations in the fit quality can be attributed primarily to geometric and physical properties of the system, rather than degeneracies between limb darkening and transit shape.

3.2.2 Initial Fit: Timing Alignment and Statistical Assessment

In the initial fitting stage, the majority of system parameters were fixed at their literature values. Only the mid-transit time, T_{centre}, was allowed to vary. This step ensured correct phase alignment between the synthetic transit model and the observed TESS light curve.

Although this initial fit successfully aligned the transit timing, the model transit appeared slightly deeper and narrower than the observed transit. In addition, the residuals exhibited systematic structure, particularly during ingress, indicating that the baseline model was not fully capturing the detailed morphology of the observed light curve. This behaviour is illustrated in **Figure 10**, which shows the baseline limb-darkened model overlaid on the observational data before parameter optimisation.

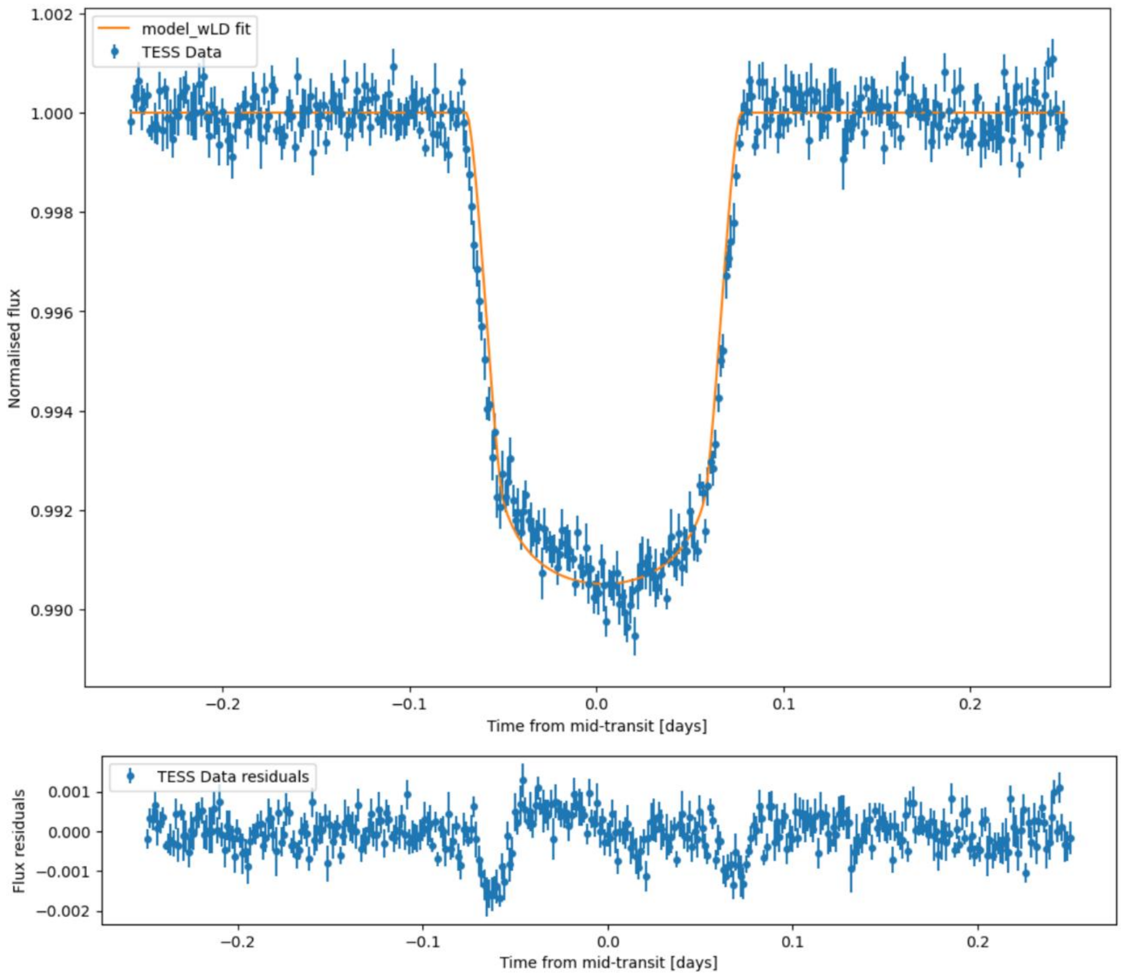

Figure 10. Observed and synthetic transit light curves of HAT-P-70. The **top panel** shows the TESS photometric data (blue points) together with the best-fit synthetic transit model including limb darkening only (orange curve). All remaining system parameters were held fixed at their published literature values. While the overall transit shape is well reproduced, systematic deviations are evident during the ingress and egress phases. The **bottom panel** displays the residuals between the TESS data and the limb-darkened model, revealing correlated structure around ingress and egress that is not captured by a standard symmetric transit model.

The quality of the initial fit was quantified using a chi-squared analysis. The resulting statistical metrics were:

- Mean photometric uncertainty: 3.60×10^{-4}
- Standard deviation of residuals: 4.92×10^{-4}
- Number of fitted parameters: 4
- Chi-squared: $\chi^2 = 816.19$
- Degrees of freedom: 356
- Reduced chi-squared: $\chi^2_\nu = 2.29$

The elevated value of χ_ν^2 demonstrates that the model does not reproduce the data within the quoted uncertainties. Furthermore, the fact that the residual scatter exceeds the mean photometric uncertainty indicates the presence of additional structure in the light curve that is not accounted for by a simple limb-darkened transit model.

These results suggest that the observed asymmetry is not purely noise-driven and that further refinement of the model is required to capture the true transit morphology.

3.2.3 Refined Fit: Allowing Key Parameters to Vary

To improve the fit, several key geometric parameters were subsequently allowed to vary. These included the planet-to-star radius ratio, $R_p/R_\star$, the scaled semi-major axis, $a/R_\star$, and the impact parameter, b. Allowing these parameters to vary led to a substantial improvement in the agreement between the model and the data. Specifically, $R_p/R_\star$decreased from 0.099 to 0.093, producing a shallower and more realistic transit depth. The scaled semi-major axis shifted from 5.45 to 5.24, slightly increasing the transit duration, while the impact parameter decreased modestly from -0.63 to -0.61, further improving the overall transit morphology.

With these updated parameters, the standard deviation of the residuals decreased significantly and became much closer to the expected noise level of the data. The reduced chi-squared improved from 2.29 to 1.55. Although still above unity, this represents a considerable improvement and indicates that the revised model provides a more accurate representation of the observed transit shape. The improved fit is shown in **Figure 11**, where the optimised model more closely follows the observed light curve.

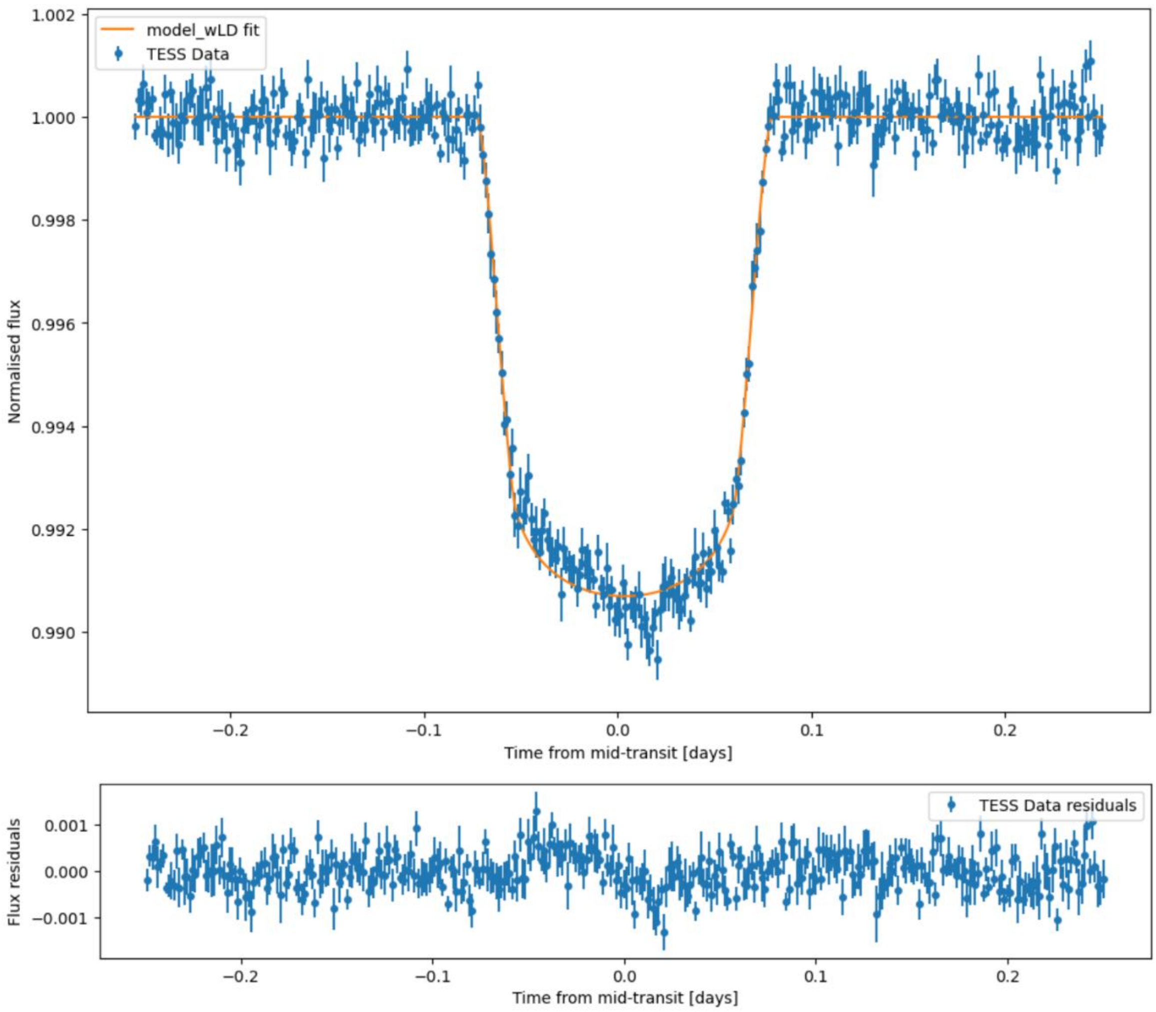

Figure 11. Observed and synthetic transit light curves of HAT-P-70 with fitted geometric parameters. The **top panel** shows the TESS photometric data (blue points) together with the best-fit synthetic transit model including limb darkening (orange curve). In this case, the key transit geometry parameters — $R_p/R_\star$, $a/R_\star$ and b— were allowed to vary during the fit, while all remaining parameters were fixed to their published literature values. The residuals are also reduced, indicating an improved fit; however, the **bottom panel** still reveals systematic deviations, particularly during the ingress and egress phases, suggesting that additional physical effects are not captured by a purely limb-darkened, symmetric transit model.

Importantly, the fitting procedure returned a very low probability of 1.45×10^{-10} that such an improvement could arise by chance. This strongly suggests that the observed asymmetry is astrophysical in origin rather than the result of random noise. The persistence of residual structure even after geometric optimisation provides compelling motivation for the introduction of additional physical effects—most notably gravity darkening—which are explored in the following sections.

3.3 Stellar Oblateness and Gravity-Darkening Framework

As established in **Chapter 1**, gravity darkening arises from the rotational deformation of the stellar surface and the resulting latitude-dependent temperature distribution (*von Zeipel, 1924; Espinosa Lara & Rieutord, 2011*). Accurate modelling of gravity-darkened transit signals, therefore, requires a physically consistent description of the stellar rotation, oblateness, and surface temperature structure. In this section, a self-consistent framework is developed to derive these quantities for the HAT-P-70 system, forming the physical foundation for the transit modelling presented in the following sections.

3.3.1 Rotational Geometry and Stellar Oblateness

To continue the analysis, it is necessary to estimate the stellar oblateness *f*, which describes the degree to which the equatorial radius exceeds the polar radius (*Dholakia et al., 2022*)

$$f = 1 - \frac{R_{\rm pole}}{R_{\rm eq}}. \qquad \textbf{Equation 3}$$

Although this quantity is sometimes quoted in the literature, it is not directly measured for most stars, including HAT-P-70, and must therefore be derived from observable rotational parameters. To achieve this, a dedicated Python notebook was developed following the approach of Dholakia et al. (2022), applying a series of rotational relationships to convert the observed projected rotational velocity into the physical parameters required for gravity-darkened transit modelling used in the next chapters.

The analysis begins with the measured projected rotational velocity $v\sin i_\star$, where $i_\star$ is the inclination of the stellar spin axis to the line of sight. The true equatorial rotation speed is given by

$$v_{\rm eq} = \frac{v\sin\, i_\star}{\sin i_\star}. \qquad \textbf{Equation 4}$$

When the star is viewed equator-on $i_\star = 90°$, the observed value corresponds directly to the equatorial velocity and represents the minimum physically allowed rotation speed for the system. As the viewing geometry approaches pole-on, $i_\star = 0°$, the inferred equatorial velocity increases, producing progressively larger rotational deformation. By scanning the full range of physically allowed inclinations, a corresponding range of equatorial velocities and stellar flattening is obtained.

From the equatorial velocity, the stellar rotation period follows directly,

$$P_{\rm rot} = \frac{2\pi R_\star}{v_{\rm eq}}, \qquad \textbf{Equation 5}$$

and the angular velocity is then

$$\Omega = \frac{2\pi}{P_{\text{rot}}}. \qquad \textbf{Equation 6}$$

For convenience, the rotation rate is expressed in dimensionless form as

$$\omega = \Omega\sqrt{\frac{R_\star^3}{GM_\star}}, \qquad \textbf{Equation 7}$$

With $0 \leq \omega < 1$, where $\omega = 1$ corresponds to the critical break-up rotation.

Using the Maclaurin-based relation adopted by Dholakia et al. (2022), the stellar oblateness is then obtained from

$$f = 1 - \frac{2}{\omega^2 + 2}, \qquad \textbf{Equation 8}$$

which naturally constrains the physically allowed range to $0 \leq f < 1/3$. Values approaching this upper limit would imply extreme rotational deformation and the onset of centrifugal instability (*Dholakia et al., 2022*).

The Python notebook developed for this project automates the entire procedure, generating the allowed ranges of $i_\star$, v_{eq}, ω, and f, and propagating these into the surface temperature calculations required for gravity-darkened transit modelling in STARRY. This provides a complete rotational and geometric characterisation of HAT-P-70 even in the absence of a directly measured oblateness.

3.3.2 Rotation-Dependent Gravity Darkening

Until this point, gravity darkening has been described using the classical von Zeipel prescription in **Chapter 1.1**, in which the local effective temperature scales with the effective surface gravity, appropriate for stars with radiative envelopes under slow rotation (*von Zeipel, 1924*). However, this approximation breaks down for rapidly rotating, oblate stars such as HAT-P-70. As the stellar rotation rate increases, the equatorial regions cool and the gravity–temperature relation weakens, reducing the effective gravity-darkening response.

Espinosa Lara & Rieutord (2011), with **Figure 12**, demonstrated that the gravity-darkening exponent is not constant but decreases systematically with increasing stellar flattening. Their analysis shows that slowly rotating stars remain close to the classical value $\beta \approx 0.25$, while fast rotators occupy substantially lower values (*Espinosa Lara & Rieutord*, *2011*). This behaviour is physically essential for reproducing the surface temperature distribution of rapidly rotating stars.

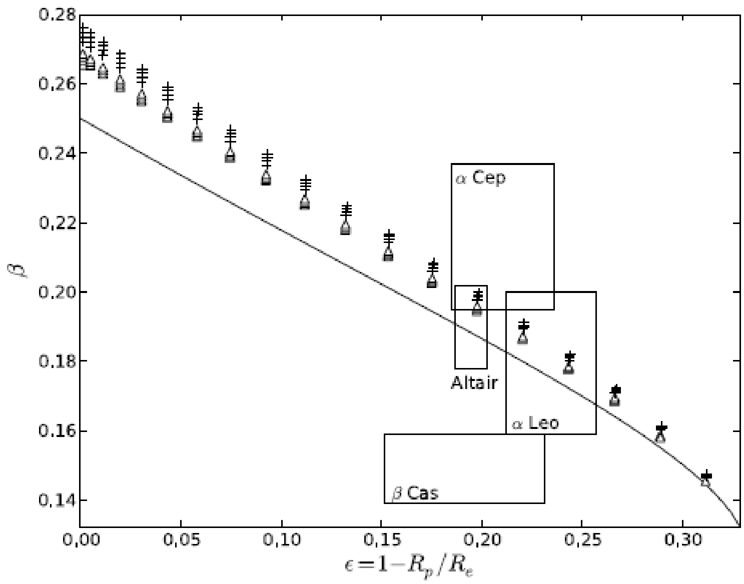

Figure 12. Variation of the gravity-darkening exponent β with stellar flattening. The plotted values are obtained by fitting the effective temperature distribution using a least-squares approach. The continuous curve shows the model introduced in this study. The boxed data points indicate empirically determined values of β, along with their uncertainties, derived from interferometric observations of several rapidly rotating stars. (Adapted from *Espinosa Lara & Rieutord, 2011*)

For this reason, the fixed von Zeipel coefficient is replaced in this work with the rotation-dependent formulation, $T_{\rm eff}(\theta) \propto g_{\rm eff}(\theta)^{\beta(f)}$, where the gravity-darkening exponent β is determined self-consistently from the stellar oblateness. Using the oblateness values computed in **Chapter 3.3.1**, β is updated for each stellar configuration, ensuring that the resulting surface temperature field remains physically consistent with the underlying stellar rotation.

3.3.3 Derived Stellar Properties of HAT-P-70

Applying the above framework to HAT-P-70 yields the stellar parameters listed in **Table 2**. The allowed stellar inclination range extends from 13.12° to 166.88°, with the corresponding values of oblateness f, dimensionless angular velocity ω, gravity-darkening exponent β, and equatorial and polar temperatures evolving systematically with viewing geometry.

$i_\star$ (degrees)	Oblateness f	Dimensionless Ang. Vel. ω	Beta β	$T_{equator}$ (K)	T_{pole} (K)
15	0.2754	0.8718	0.16	8023.69	8898.96
25	0.1254	0.5355	0.21	8215.78	8690.9
35	0.0724	0.3950	0.23	8307.55	8594.9
45	0.0489	0.3206	0.23	8351.63	8549.53
55	0.0369	0.2769	0.24	8374.87	8525.8
65	0.0304	0.2503	0.24	8387.8	8512.67
75	0.0268	0.2348	0.24	8394.84	8505.52
85	0.0253	0.2276	0.24	8397.99	8502.33
90	0.0251	0.2268	0.24	8398.38	8501.94
95	0.0253	0.2276	0.24	8397.99	8502.33
105	0.0268	0.2348	0.24	8394.84	8505.52
115	0.0304	0.2503	0.24	8387.8	8512.67
125	0.0369	0.2769	0.24	8374.87	8525.8
135	0.0489	0.3206	0.24	8351.63	8549.53
145	0.0724	0.3950	0.23	8307.55	8594.9
155	0.1254	0.5355	0.22	8215.78	8690.9
165	0.2754	0.8718	0.16	8023.69	8898.96

Table 2 Derived stellar parameters — including inclination $i_\star$, oblateness f, dimensionless angular velocity ω, gravity-darkening coefficient β, and the resulting polar and equatorial temperatures (T_{pole} and T_{eq}) computed using a custom Python notebook. Values are reported over the full range of physically allowed stellar inclinations for HAT-P-70, from 13.12° to 166.88°, corresponding to the limits before rotational breakup occurs.

As the inclination approaches an equator-on configuration ($i_\star = 90°$), both f and ω reach their minimum values before increasing symmetrically toward pole-on orientations. The resulting temperature distribution follows the expected gravity-darkening relation $T \propto g^\beta$, producing hotter polar regions and cooler equatorial zones. For low inclinations, the observed stellar flux is dominated by the hot poles, leading to a larger pole-to-equator temperature contrast, while near equator-on configurations yield a more balanced temperature profile.

However, the derived temperatures and stellar oblateness do not reproduce the values reported by Cauley & Ahlers (2022) for an inclination of $i_\star = 35°$. In their study of HAT-P-70, they obtained $T_{\text{pole}} = 8670 \pm 300$ K, $T_{\text{eq}} = 7830 \pm 290$ K and $f \approx 0.08$.

A significant portion of this discrepancy arises from differences in the adopted definition of the stellar inclination. In the present work, $i_\star = 90°$ corresponds to an equator-on configuration, whereas Cauley & Ahlers (2022) define $i_\star = 0°$ as pole-on. Under their convention, an inclination of 35° therefore, corresponds to 55° in our framework. This difference in coordinate definition introduces ambiguity and can lead to incorrect values of $i_\star$ if not handled carefully.

Although translating between these conventions partially reduces the disagreement, it does not fully reconcile the results: even after adopting the equivalent geometry, the inferred oblateness and temperature contrast remain significantly smaller in the present analysis. This inconsistency indicates that a more rigorous reconciliation of coordinate conventions and underlying model assumptions is required, and this issue is addressed explicitly during the transit-modelling phase.

3.4 STARRY Simulations of Gravity-Darkened Transits

Transit light-curve simulations were generated using the STARRY Python package, originally developed by Luger et al. (2019). The package models stellar and planetary surface-brightness distributions using spherical harmonics and enables exact analytic computation of transit and occultation light curves, including the effects of limb darkening, gravity darkening, and stellar oblateness.

During the course of this project, it was established—following detailed discussions with my supervisor—that the publicly released version of STARRY is no longer actively maintained and is incompatible with several modern Python dependencies. As a result, the standard distribution could not be reliably used for the present analysis. Instead, this work employs a modified and actively maintained version of the STARRY framework curated by UCL Observatory, which preserves the original physical formalism while ensuring numerical stability, compatibility with current software libraries, and continued functionality for gravity-darkened transit modelling.

This section investigated the sensitivity of gravity-darkened transit light curves to variations in the key stellar and orbital parameters governing the geometry and surface-brightness distribution of the HAT-P-70 system. Using the STARRY framework, a suite of synthetic transit models was generated in order to assess whether physically distinct configurations produced observationally distinguishable signatures in the transit morphology, or whether such differences remained largely degenerate within the precision of the available TESS photometry.

A particularly informative test case arose from the discrepancy between the stellar parameters derived in **Chapter 3.3** and those reported by Cauley & Ahlers (2022), specifically the stellar inclination $i_\star$, the oblateness f, and the polar temperature T_{pole}. These differences provided an opportunity to evaluate how uncertainties in stellar geometry and gravity-darkening prescriptions propagated into the observable transit signal.

Two principal stellar configurations were therefore examined and reported in **Figure 13**.

1. **Configuration A** (*Cauley & Ahlers, 2022*):

   $i_\star = 35^\circ$,
   $f = 0.0800$,
   $T_{\mathrm{pole}} = 8670$ K

2. **Configuration B** derived from the rotational analysis presented in **Chapter 3.3**:

   $i_\star = 55^\circ$,
   $f = 0.0369$,
   $T_{\mathrm{pole}} = 8526$ K

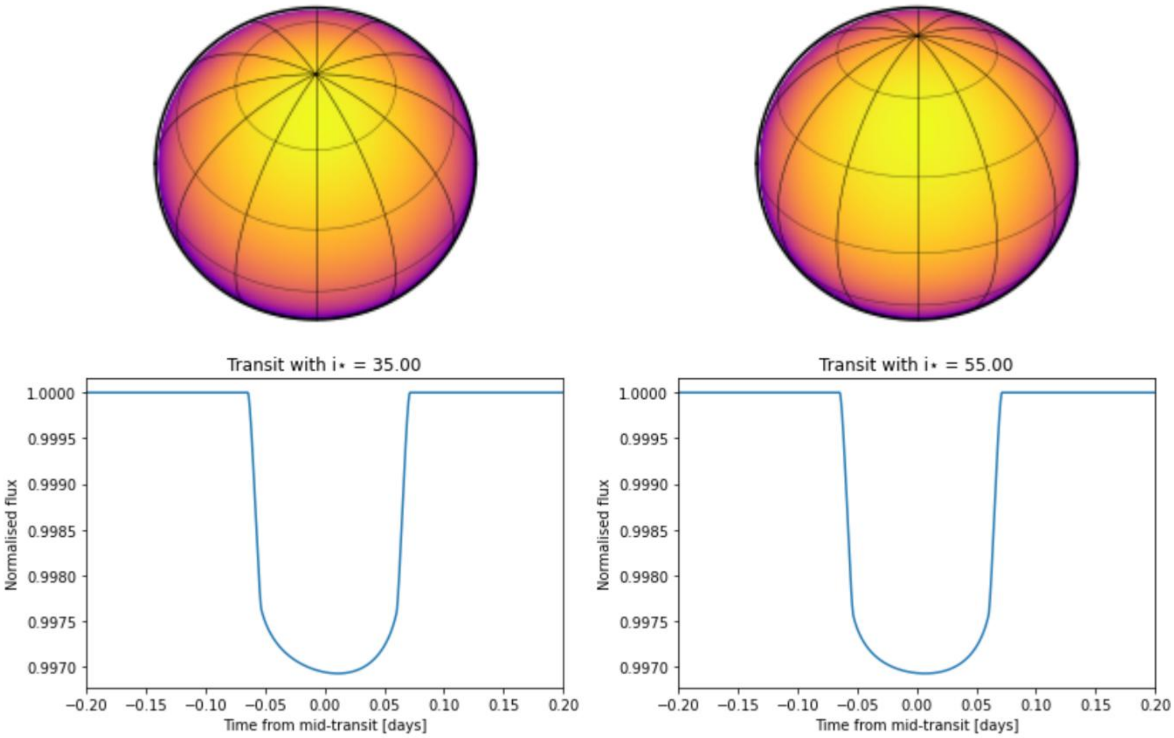

Figure 13. Synthetic light curves and system configurations for HAT-P-70 at different stellar inclinations. The left panel shows the configuration with stellar inclination $i_\star = 35^\circ$, while the right panel corresponds to $i_\star = 55^\circ$.

For each configuration, a systematic exploration was performed over the key parameters that controlled the gravity-darkened transit morphology: the sky-projected obliquity λ, the gravity-darkening exponent β, and the stellar inclination $i_\star$. The resulting synthetic light curves were then compared directly with the observed TESS light curve presented in **Chapter 3.1**, in order to evaluate which combination of parameters best reproduced the observed asymmetry.

3.4.1 Dependence on the Projected Obliquity λ

The initial set of experiments adopted **Configuration B**, using the parameters reported by Cauley & Ahlers (2022). The projected obliquity λ was varied from $45^\circ \rightarrow 90^\circ \rightarrow 113^\circ$, while all other parameters were held fixed.

These models assumed a gravity-darkening coefficient of $\beta = 0.20$, as adopted by Cauley & Ahlers (2022), which was slightly lower than the value predicted by the Espinosa Lara & Rieutord (2011) prescription for the corresponding oblateness, but was shown by those authors to provide the best fit for HAT-P-70.

The resulting synthetic transits exhibited clear and systematic changes in morphology. At $\lambda = 45^\circ$, the transit became noticeably shallower with a deeper egress, reflecting the planet's trajectory across progressively hotter stellar latitudes near the pole. For $\lambda = 90^\circ$ and 113°, the overall depth decreased further while the characteristic asymmetric shape persisted, with a shallower ingress and deeper egress. They all resulted in acceptable results.

The same analysis was repeated using **Configuration A**, with $i_\star = 55^\circ$ and the corresponding values of f and T_{pole} derived in **Chapter 3.3**. For identical values of λ, the resulting transits again displayed asymmetric shapes qualitatively similar to the observed TESS data. However, the asymmetry was less pronounced, and the overall transit depth was slightly shallower than in the $i = 35^\circ$ case.

This behaviour was physically expected: the lower oblateness, reduced pole-to-equator temperature contrast, and slightly higher gravity-darkening exponent in Configuration A naturally weakened the gravity-darkening signal. Nevertheless, the persistence of a measurable asymmetry confirmed that both configurations remained viable within the photometric precision of the data. To demonstrate that the difference was not substantial, the results were presented in **Figure 14**.

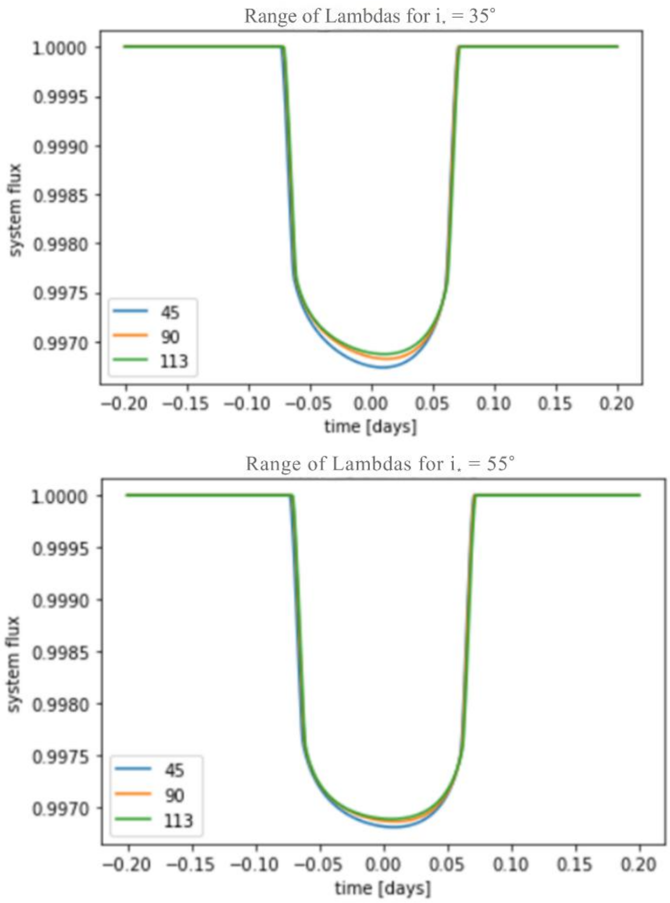

Figure 14. Synthetic transit light curves for HAT-P-70 b illustrating the effect of different spin–orbit misalignment angles (λ). The **top panel** shows the light-curve shapes obtained for a stellar inclination of $i_\star = 35^\circ$, with each colour representing a different value of λ: blue (45°), orange (90°), and green (113°), the latter corresponding to the value reported in the literature. The bottom panel presents the same set of λ values for $i_\star = 55^\circ$. In both configurations, the transit asymmetry is characterised by a shallower ingress and a deeper egress, with the effect appearing sharper and more pronounced for $i_\star = 35^\circ$.

3.4.2 Sensitivity to Stellar Inclination and Gravity-Darkening Exponent

To further probe parameter degeneracies, additional simulations were performed in which only the stellar inclination $i_\star$ was varied between 35° and 55°, with all other parameters held fixed. The resulting transit shapes exhibited only minor differences, indicating that both inclinations were compatible with the observed data and suggesting the presence of a degenerate family of solutions.

A similar test was conducted for the gravity-darkening exponent, comparing $\beta = 0.20$ and $\beta = 0.24$. Although small changes in transit morphology were observed, the differences remained subtle and did not significantly alter the quality of the fit in both cases.

3.4.3 Implications for System Geometry

Taken together, these simulations demonstrated that while variations in λ, $i_\star$, and β did influence the detailed transit morphology, the observed light curve of HAT-P-70 did not uniquely favour a single configuration. The most robust distinction was that models with $i_\star = 35°$ generally produced slightly stronger asymmetry and deeper transits than those with $i_\star = 55°$.

This behaviour motivated the statistical modelling approach developed in **Chapter 4**. Only through a full Markov Chain Monte Carlo exploration of parameter space could the true posterior distribution of $i_\star$, λ, and the resulting three-dimensional spin–orbit angle ψ be constrained. The STARRY simulations presented here therefore provided the physical and observational foundation for the inference framework applied in the following chapter, and for the final determination of the spin–orbit architecture of the HAT-P-70 system via the Masuda (2015) formalism.

3.5 Statistical Inference of the System Geometry

3.5.1 Why a Statistical Approach Was Needed

The simulations in **Chapter 3.4** showed that more than one physical configuration of the HAT-P-70 system could reproduce the observed transit asymmetry. In other words, different combinations of stellar inclination, gravity darkening and orbital orientation could all produce light curves that looked very similar within the precision of the TESS data. This meant that simply finding one "best" model was not enough: the problem is inherently uncertain, and this uncertainty needs to be quantified.

For this reason, a statistical approach was required. Rather than asking "what is the single best solution?", the goal became "which solutions are most consistent with the data, and how uncertain are they?". This way of thinking follows the Bayesian view of data analysis, in which our knowledge of the parameters is updated gradually as information from the data is taken into account (*Brewer, 2014*)

3.5.2 Overview of the MCMC Method and Chosen Parameters

The statistical analysis was carried out using a custom Markov Chain Monte Carlo (MCMC) framework developed by the project supervisor. The purpose of the MCMC was to explore a large range of physically possible system configurations and identify which of them produced gravity-darkened transit models that best matched the observed TESS light curve.

The set of parameters explored in the simulations was $\theta = (i_\star, f, \lambda, R_\star, M_\star, i_{\mathrm{orb}}, a/R_\star)$ corresponding to the stellar inclination, stellar oblateness, projected spin–orbit angle, stellar radius, stellar mass, orbital inclination, and the scaled semi-major axis. Together, these quantities fully helped describe the three-dimensional geometry of the system and the shape of the gravity-darkened transit.

At each step of the simulation, a new combination of these parameters was proposed. For that combination, a synthetic gravity-darkened transit light curve was generated using the STARRY framework and compared directly with the observed TESS photometry. The photometric component of the likelihood was computed from the residuals between the model and the observed flux values, assuming Gaussian-distributed uncertainties.

In addition to the transit photometry, Gaussian constraint terms were included in the likelihood for the projected stellar rotation velocity $v\sin i_\star$ and the stellar surface gravity log g, using literature values and their associated uncertainties. These terms penalised solutions that reproduced the transit morphology but were inconsistent with independent spectroscopic measurements. The total likelihood was therefore defined as the product of the photometric likelihood and the stellar constraint terms, following the Bayesian formalism described by Brewer (2014).

If a proposed parameter set resulted in a higher likelihood, it was preferentially retained by the chain; if the likelihood was lower, the proposal was accepted with a probability determined by the likelihood ratio. Repeating this process many thousands of times allowed the MCMC to map out the regions of parameter space most consistent with both the TESS light curve and external stellar constraints.

This approach reflects the core principle of Bayesian inference, in which model parameters are treated as probability distributions rather than fixed unknown quantities, and are updated iteratively as information from the data is incorporated (*Brewer, 2014*).

Uniform prior ranges were adopted for the stellar inclination $i_\star$ and stellar oblateness f, in order to avoid imposing strong assumptions on these quantities. The remaining parameters were allowed to vary within physically reasonable bounds and within the uncertainties reported in the literature. This ensured that the analysis remained physically realistic while allowing the observational data to guide the final inferred system geometry.

3.5.3 Exploring Two Physical Configurations

As established in **Chapter 3.4**, two physically motivated stellar configurations remained viable: Configuration A from the literature (*Cauley & Ahlers, 2022*) and Configuration B derived from the rotational framework of this work. The full MCMC analysis was therefore performed independently for both configurations.

For each case, the simulation was run for many thousands of steps. As the chains evolved, they gradually concentrated around regions of parameter space that produced the best agreement between the model and the TESS light curve. This behaviour—where the simulation spends more time exploring good solutions and less time on poor ones—is the defining feature of MCMC methods and allows complicated, multi-dimensional problems to be tackled efficiently (*Brewer, 2014*)

If both initial configurations converged toward similar posterior solutions, the resulting geometry could be regarded as robust and largely independent of the assumed starting conditions.

3.5.4 Validation of the Inference and Determination of the System Geometry

Once the simulations had stabilised and the initial transient phase had been removed, the resulting parameter samples were examined using the *corner* plotting library in Python (*Foreman-Mackey, 2016*). These plots provided an intuitive visual summary of the results, showing both the uncertainty on each parameter and the correlations between different parameters.

Stable, well-defined distributions indicated that the simulation had converged and that the inferred parameter values were reliable. This visual inspection formed an essential part of the analysis, as recommended in practical Bayesian workflows (*Brewer, 2014*).

Finally, the posterior distributions of the orbital inclination i_{orb}, stellar inclination $i_\star$, and projected obliquity λ were combined to compute the true three-dimensional spin–orbit angle ψ.

4 Results and Discussion

This chapter presents the results of the gravity-darkened transit modelling and the associated Markov Chain Monte Carlo (MCMC) analysis applied to the TESS light curve of HAT-P-70. Building on the theoretical framework and modelling methodology developed in **Chapter 3**, the goal here is to assess whether gravity-darkened transit photometry can self-consistently reproduce the observed transit morphology and constrain the three-dimensional spin–orbit geometry of the system.

In **Chapter 3.2**, conventional limb-darkened transit models were shown to provide an inadequate description of the observed light curve. Although allowing key geometric parameters to vary significantly improved the fit, statistically significant residual structure remained, particularly during ingress, egress, and near mid-transit. A chi-squared analysis demonstrated that the probability of the observed transit morphology arising by chance was extremely low, strongly suggesting the presence of additional astrophysical effects beyond standard symmetric transit models.

One particularly notable feature of the data is a subtle but persistent structure near mid-transit, visible as a localised deviation in the residuals. This feature cannot be explained by limb darkening, timing offsets, or simple geometric adjustments. During the course of this analysis, the origin of the residual structure near mid-transit was discussed in detail with the project supervisor. Several alternative astrophysical explanations were considered in addition to gravity darkening. These included the presence of stellar surface inhomogeneities such as starspots or bright regions (*Balona, 2017; Balona, 2021*) as well as stellar pulsations, in particular δ Scuti–type variability, which is commonly observed in rapidly rotating A-type stars (*Bowman & Kurtz, 2018*). Each of these mechanisms can, in principle, introduce localised distortions in transit light curves that may mimic or contaminate gravity-darkening signatures.

However, a rigorous investigation of these possibilities would have required either multi-band photometry, higher-cadence observations, or an extended time baseline to identify repeatable variability patterns. Such an analysis lay beyond the scope of the present work. It was anticipated that the release of additional TESS observations from Sector 98 would provide improved photometric coverage and enable a more detailed characterisation of the transit morphology. At the time of completing this thesis, however, the Sector 98 data products had not yet been publicly released, preventing further refinement of the light curve model. As a result, the unexplained mid-transit structure was not conclusively resolved within this study.

Motivated by the system's extreme stellar rotation and strong theoretical expectation for gravity darkening, the modelling effort was extended to fully gravity-darkened transit simulations coupled with Bayesian inference. Two physically motivated stellar configurations—derived from the literature and from the rotational analysis in **Chapter 3**—were adopted as initial starting points to test the robustness of the inferred system geometry and to investigate discrepancies between previous studies.

4.1 Light Curve Fits and MCMC Exploration

The MCMC analysis was initiated with the aim of determining whether the observed asymmetric transit morphology of HAT-P-70b could be reproduced by gravity-darkened models and, if so, whether the inferred system geometry remained stable under different initial assumptions. A further motivation was the presence of two apparently inconsistent stellar inclination values reported in the literature, which raised questions about how uniquely the geometry of the system could be constrained.

Cauley & Ahlers (2022) reported a relatively low stellar inclination, while Zhou et al. (2019) inferred a value closer to $i_\star \approx 55^\circ$, in agreement with the rotational framework developed in Chapter 3 of this work. As discussed previously, a substantial part of this discrepancy may arise from differences in the adopted definition of stellar inclination. In particular, it appears that Cauley & Ahlers (2022) may have adopted a convention in which $i_\star = 0^\circ$ corresponds to an equator-on configuration, whereas in this work, $i_\star = 90^\circ$ represents an equator-on star. This interpretation was inferred from their reported geometry and modelling choices, although the inclination convention is not stated explicitly in their paper. As a result, direct comparison between published values is non-trivial and motivated a careful, data-driven reassessment of the system geometry.

To explore this issue, the MCMC simulations were initialised using alternative, physically motivated stellar configurations consistent with both the literature and the rotational and gravity-darkening framework developed in **Chapter 3**. Rather than enforcing a preferred solution a priori, the analysis was designed to test whether the gravity-darkened transit modelling naturally converged toward a consistent geometric configuration or whether the inferred parameters remained sensitive to the adopted starting assumptions.

For all simulations, most system parameters were initialised close to their literature values. The stellar inclination $i_\star$ and stellar oblateness f, however, were allowed to vary freely under uniform priors. This choice avoided imposing strong assumptions on the stellar geometry and allowed the MCMC chains to explore the physically allowed parameter space and identify configurations favoured directly by the data.

While part of the motivation for this analysis was to assess whether the parameters recovered in this work were consistent with previously published results, the primary objective was to derive an independent and self-consistent set of geometric parameters from the gravity-darkened transit modelling. These recovered parameters were then used to characterise the three-dimensional architecture of the HAT-P-70 system and to compute the true spin–orbit angle ψ. Any agreement or discrepancy with literature values was subsequently evaluated and interpreted in the context of the modelling assumptions and physical framework adopted, as discussed in the following sections.

4.1.1 Configuration A: Behaviour and Failure to Converge

After approximately 100,000 iterations, the MCMC chain initialised under **Configuration A** began to show behaviour that was difficult to reconcile with both the data and prior expectations for the HAT-P-70 system. Most notably, the inferred stellar inclination $i_\star$ drifted steadily towards high values, frequently exceeding 70°. While such inclinations are not forbidden by break-up constraints, they fall well outside the range reported in the literature for HAT-P-70 and beyond the parameter space motivated in **Chapter 3**. This gradual migration suggested that the sampler was moving away from physically plausible configurations rather than converging towards a solution supported by existing observational constraints.

This impression was reinforced by the trace diagnostics. Rather than settling into a stable region of parameter space, the chains continued to wander, particularly in $\cos i_\star$ and the corresponding inclination. Even after transforming back to $i_\star$, the posterior distribution remained broad and poorly defined. In practice, this indicated that the likelihood surface under Configuration A did not sufficiently penalise unphysical geometries, allowing the sampler to explore a wide range of inclinations without identifying a preferred solution.

The consequences of this behaviour became evident upon inspection of the model light curves generated during the early stages of the MCMC chain. Despite the inclusion of gravity darkening, the simulated transits produced during the first ∼100,000 iterations were largely symmetric and closely resembled conventional limb-darkened transit profiles. Consequently, the asymmetric ingress–egress morphology observed in the TESS data, as well as the subtle structure near mid-transit, were not reproduced by the model.

This qualitative discrepancy was confirmed quantitatively through a χ^2 analysis. The reduced chi-squared value for the in-transit region was very large ($\chi^2_{\rm red} = 2.25$), while the out-of-transit portion also yielded a substantial value ($\chi^2_{\rm red} = 1.37$), indicating that the model provided a statistically unacceptable fit to the data.

Further evidence of the model's inadequacy is visible in the residuals. Rather than exhibiting reduced scatter consistent with a successful fit, the residuals retained strong correlated structure, particularly during ingress and egress (**Figure 15**). This behaviour is consistent with the model failing to capture the intrinsic asymmetry of the observed light curve and was likely compounded by the unresolved mid-transit feature discussed earlier.

To investigate whether this structure could be attributed to underestimated uncertainties or systematic errors, an additional analysis was performed in which the in-transit reduced χ^2 was rescaled by the square root of the out-of-transit reduced χ^2. Although this adjustment reduced the in-transit value to $\chi^2_{\rm red} = 1.67$, it remained sufficiently large to indicate a persistent mismatch between the model and the data. Rather than removing correlated structure from the residuals, the gravity-darkened model under Configuration A was effectively reduced to a symmetric approximation, failing to exploit the asymmetric signatures expected from gravity darkening.

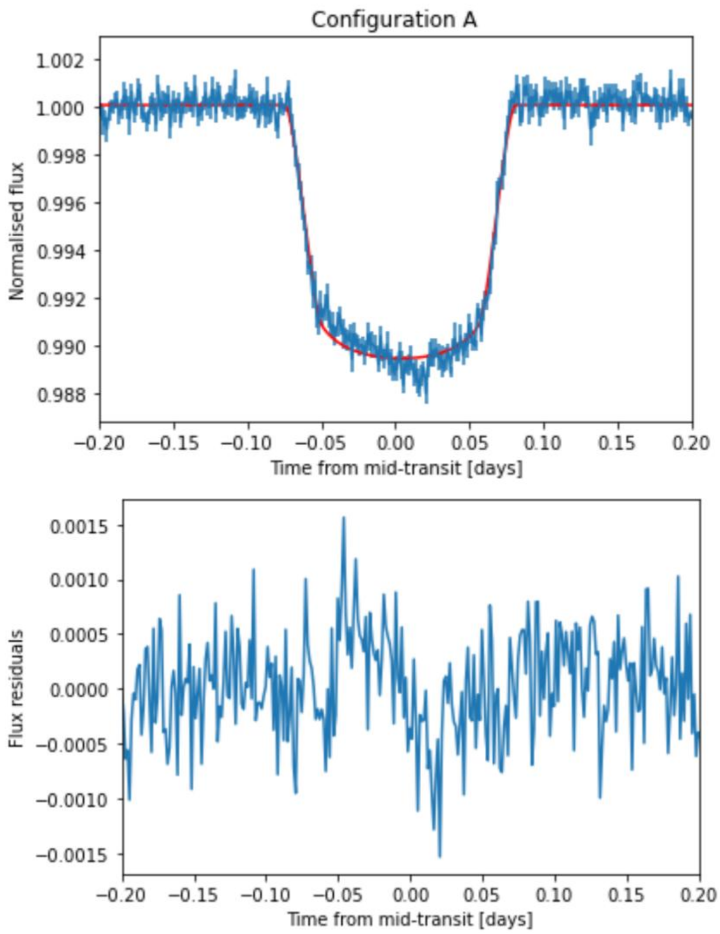

Figure 15. Synthetic gravity-darkened transit light curve for the HAT-P-70 system obtained using **Configuration A**. The red curve shows the best-fitting gravity-darkened transit model, while the blue points represent the phase-folded TESS photometric data with associated uncertainties. The **top panel** displays the observed transit shape together with the model fit, whereas the **bottom panel** shows the residuals between the data and the model. Significant correlated structure remains in the residuals, particularly during the ingress and egress phases, indicating that this configuration does not fully reproduce the observed transit morphology.

Although the stellar inclination and oblateness were allowed to vary freely, the sampler did not evolve towards configurations capable of producing a measurable gravity-darkening signal. The stellar surface maps inferred from the posterior samples remained only weakly asymmetric, resulting in transit chords that sampled nearly symmetric brightness distributions. This provides a natural explanation for why the resulting light curves failed to exhibit the expected asymmetries: under Configuration A, the model simply did not access regions of parameter space where gravity darkening meaningfully influenced the transit shape.

Further insight was provided by the corner plots. Rather than converging towards well-defined, approximately Gaussian posteriors, several of the directly sampled geometric parameters—most notably the stellar inclination $i_\star$ and the projected spin–orbit angle λ—exhibited broad, irregular, and in some cases multi-modal distributions. While correlations between parameters were present, they did not collapse towards a physically interpretable solution. Instead, the chains continued to explore disparate regions of parameter space, indicating that no stable or self-consistent configuration was being identified for the parameters required to compute the true spin–orbit angle.

At this stage, the true spin–orbit angle ψ was not investigated directly. The purpose of the MCMC analysis was to obtain reliable posteriors for the individual geometric parameters $i_\star$, λ, and i_{orb}, which would later be combined using the Masuda relation to infer ψ. Since these foundational parameters failed to converge under Configuration A, any derived constraint on ψ would be physically meaningless. For this reason, the analysis was halted at this stage and Configuration A was deemed unsuitable for further interpretation.

4.1.2 Configuration B: Initial Success and Motivation for Extended Runs

In parallel with Configuration A, an identical MCMC analysis was performed using **Configuration B** as the initial starting point. After a comparable number of iterations (approximately 150,000), the behaviour of the MCMC chains differed markedly from that observed previously. Several key parameters—most notably the projected spin–orbit obliquity λ, the orbital inclination i_{orb}, and the stellar mass and radius—began to converge toward well-defined values. This improvement suggested that, under Configuration B, the gravity-darkened transit model was starting to capture important features of the observed transit light curve.

Despite this encouraging behaviour, not all parameters exhibited satisfactory convergence. In particular, the stellar inclination $i_\star$ and the stellar oblateness f remained poorly constrained throughout the simulations, showing no clear signs of convergence even after the full set of iterations. While the corresponding model light curves demonstrated a noticeable improvement compared to Configuration A—most prominently through the emergence of transit asymmetries consistent with gravity darkening—the overall quality of the fit remained unsatisfactory. Both the chi-squared and reduced chi-squared values remained high, particularly during the in-transit and out-of-transit phases. Furthermore, the mid-transit structure identified in **Chapter 3** was still not adequately reproduced by the model, indicating that some relevant physical processes were not yet fully captured.

The acceptance rate of the MCMC chain was relatively high, remaining close to 0.50, which indicated that the sampling itself was stable and efficient. This suggested that the lack of convergence for certain parameters was unlikely to be caused by numerical instabilities or poor chain mixing. Instead, attention was directed toward the choice of proposal distributions, particularly for $i_\star$ and f. In Configuration B, the jump sizes for these parameters were adopted directly from uncertainties reported in the literature, primarily from Cauley & Ahlers (2022). Specifically, the stellar inclination was assigned an uncertainty of $\pm 11^\circ$, while the oblateness was allowed to vary with an uncertainty of approximately 0.04 around a central value of $f \approx 0.08$. These relatively large uncertainties resulted in broad proposal distributions, which may have hindered efficient exploration of a highly correlated parameter space.

Given the strong degeneracies between stellar inclination, stellar oblateness, and gravity-darkening parameters, it was hypothesised that the absence of convergence was driven primarily by excessively large jump sizes rather than by a fundamental mismatch between the model and the data. Large proposal steps can cause the sampler to move too aggressively through correlated regions of parameter space, preventing the chains from settling into stable posterior distributions (*Brewer, 2014*)

To address this issue, two corrective steps were undertaken. First, the total number of MCMC iterations was substantially increased. An additional 400,000 iterations were performed, with the initial 100,000 steps treated as burn-in, in order to assess whether convergence might emerge at later stages once the chains had more thoroughly explored the parameter space. Second, a third configuration—hereafter referred to as **Configuration C**—was introduced. Configuration C retained the same central parameter values as Configuration B but adopted significantly reduced jump sizes for the most problematic parameters. In particular, the uncertainty on the stellar oblateness was reduced from 0.04 to 0.01, and the uncertainty on the stellar inclination was reduced from 11° to 1°. This modification was designed to test whether more conservative proposal distributions would allow the sampler to more effectively explore the correlated parameter space and achieve convergence for $i_\star$ and f.

After a total of 405,000 samples, with 101,250 discarded as burn-in, clear signs of convergence began to emerge for a large subset of parameters. In particular, the stellar radius $R_\star$, stellar mass $M_\star$, planet-to-star radius ratio $R_p/R_\star$, orbital inclination i_{orb}, projected obliquity λ, mid-transit time t_0, and flux offset y_c all converged toward well-defined regions of parameter space. The corresponding corner plots showed a single, well-defined central peak consistent with approximately normal posterior distributions. The acceptance rate remained close to 0.50, further confirming the robustness of the sampling.

However, the behaviour of the stellar inclination $i_\star$ and the oblateness f remained notably different. The stellar inclination showed partial convergence, stabilising around a value of approximately 38°, with signs of convergence becoming apparent within the final 150,000 samples. In contrast, the oblateness parameter did not exhibit similar stabilisation. Instead, it continued to drift even after a large number of iterations. The corner plots revealed the presence of two distinct peaks in the posterior distribution of f, and $i_\star$ which were interpreted as evidence for a bimodal solution. Such behaviour is consistent with the strong degeneracies inherent in gravity-darkened transit modelling and suggests that multiple physically plausible configurations may reproduce the observed asymmetry.

The best-fitting model light curves exhibited the characteristic asymmetric transit morphology expected for a gravity-darkened stellar host, lending support to the original hypothesis. Nevertheless, when applying the same statistical analysis used for Configuration A, the chi-squared and reduced chi-squared values, although improved, remained moderately high ($\chi^2_{red} = 1.36$).

These remaining discrepancies were again driven primarily by structured residuals near mid-transit, approximately within the range $0.00 < x < 0.05$. This behaviour is also apparent in the residuals plot, which does not exhibit a smooth, noise-dominated distribution. Instead, pronounced, systematic deviations persist, with notable spikes during ingress and egress (**Figure 16**).

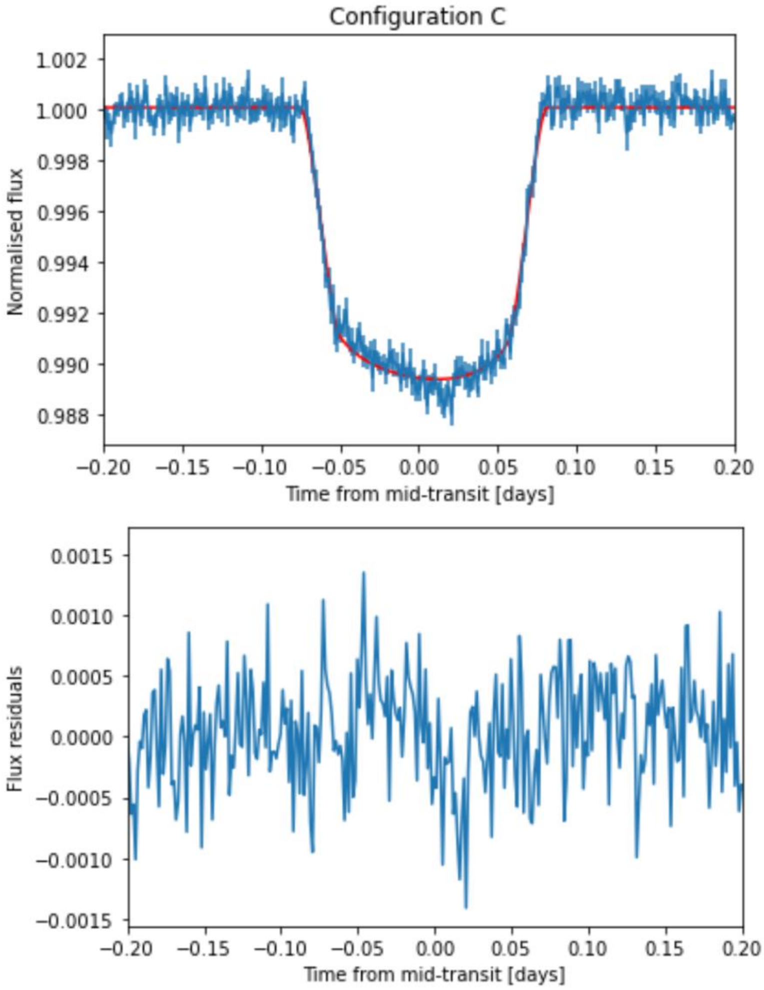

Figure 16. Synthetic gravity-darkened transit light curve for the HAT-P-70 system obtained using **Configuration C**. The red curve shows the gravity-darkened transit model, while the blue points represent the phase-folded TESS photometric data with associated uncertainties. The **top panel** displays the observed transit shape together with the model, revealing a more pronounced asymmetry relative to Configuration A. The **bottom panel** shows the residuals between the data and the model, which are generally flatter overall, indicating an improved fit. Nevertheless, significant residual structure remains during the ingress and egress phases.

A more detailed investigation into the origin of this structure is therefore required. Several physical explanations may contribute to these residuals, including stellar spots, bright regions, or pulsational variability such as δ Scuti oscillations, all of which are plausible for a hot, rapidly rotating host star. Additional photometric data—particularly from TESS Sector 98—would have been invaluable in further constraining these effects and refining the model. However, at the time of writing, Sector 98 data had not yet been released. Consequently, a deeper exploration of these potential mechanisms was beyond the scope of this project.

Despite these limitations, the results presented here strongly motivate continued observational and modelling efforts for this system. Further high-precision photometry and extended gravity-darkened modelling could provide critical insight into the stellar properties and dynamical architecture of this particularly intriguing target, HAT-P-70, which remains an excellent candidate for future follow-up studies.

4.2 Adopted Parameters and Three-Dimensional System Geometry

Following the Markov Chain Monte Carlo (MCMC) analysis described in **Chapters 4.1.1** and **4.1.2**, a single physically consistent solution was adopted to describe the HAT-P-70 system. The final system parameters selected from the posterior distributions obtained using gravity-darkened transit modelling are presented in **Table 3**. For each parameter, the quoted value corresponds to the posterior mean, with the associated uncertainties derived from the corresponding credible intervals.

Parameters	This work	Cauley & Ahlers (2022)	Zhou et al. (2019)
Equatorial Radius ($R_\odot$)	2.077 ± 0.011	2.17 ± 0.02	
Star Mass ($M_\odot$)	1.900 ± 0.011	$1.858^{+0.119}_{-0.091}$	$1.858^{+0.119}_{-0.091}$
Oblateness (f)	0.071 ± 0.002	$0.08^{+0.04}_{-0.03}$	
$i_\star$ (degree)	38.75 ± 0.50	35^{+11}_{-8}	$58.8^{+7.5}_{-4.8}$
$R_p/R_\star$	0.100		$0.099^{+0.001}_{-0.000}$
i_{orb} (degree)	97.60 ± 0.12	96.5 ± 1.2	$96.5^{+1.42}_{-0.91}$
λ (degree)	105.2 ± 3.6	113 ± 11	$113.1^{+5.1}_{-3.4}$
$v \sin i$ (km/s)	100.12	99.9 ± 0.6	$99.85^{+0.64}_{-0.61}$
Planet Radius ($R_\odot$)	0.208	$0.188^{+0.010}_{-0.015}$	$0.188^{+0.010}_{-0.015}$
Semi-major Axis a (AU)	0.047	0.047	0.047
$a/R_\star$	4.92		$5.45^{+0.29}_{-0.49}$
Gravity Darkening (β)	0.23	0.20	0.24
ψ (degree)	105.26		
t_0	0.0025		
T_{pole} (K)	8816	8670 ± 300	
$T_{equator}$ (K)	8242	7830 ± 290	

Table 3. Stellar and planetary parameters obtained from the gravity-darkened transit analysis performed in this project. The adopted values correspond to the posterior means derived from the Markov Chain Monte Carlo simulations, with uncertainties representing the associated credible intervals. Where available, these parameters are compared with values reported by Cauley & Ahlers (2022) and Zhou et al. (2019) for the HAT-P-70 system.

A comparison with previously published results revealed that, while several parameters were broadly consistent with values reported in the literature, notable discrepancies on others were present when compared with the analyses of Cauley & Ahlers (2022) and Zhou et al. (2019). These differences provided important insight into both the sensitivity of gravity-darkened transit modelling and the underlying assumptions adopted in earlier studies.

Beginning with the stellar radius, the value obtained in this work was larger than those reported by both Cauley & Ahlers (2022) and Zhou et al. (2019). This difference was interpreted as a direct consequence of the stellar radius definition adopted in the present analysis. In the gravity-darkened framework employed here, the stellar radius corresponded to the equatorial radius of the oblate star, rather than a spherical or mean radius typically assumed in standard transit analyses. Because rapid rotation increases the equatorial radius relative to the polar radius, a larger stellar radius was expected in this context. This interpretation is physically consistent with the oblate geometry inferred for HAT-P-70 and does not necessarily indicate a fundamental disagreement with earlier studies.

The stellar mass derived in this work was found to be slightly larger than literature values, as expected. This is a direct consequence of adopting a larger stellar radius—specifically the equatorial radius—within the model. The increased radius naturally leads to a higher inferred stellar mass and propagates into several related parameters, including $R_p/R_\star$ and $a/R_\star$ and, as well as the derived stellar surface gravity. As a result, these parameters differ slightly from previously reported values but remain largely consistent with the literature within their respective uncertainties (*Zhou et al., 2019; Cauley & Ahlers, 2022*). Several additional parameters, including the orbital inclination and projected stellar rotational velocity, also show good agreement with published results. This overall consistency supports the conclusion that the global system geometry recovered by the MCMC analysis remains robust and compatible with existing spectroscopic and photometric constraints.

More unexpected results were obtained for the projected spin orbit angle, the stellar oblateness and stellar inclination. As discussed in **Chapter 3**, it had initially been argued that the stellar inclination reported by Cauley & Ahlers (2022), quoted as approximately 35°, would correspond to an inclination of approximately 55° under the angular convention adopted in this work, where an inclination of 90° represents an equator-on configuration. Based on this reasoning, the expectation at the outset of the analysis was that the posterior distributions would favour a solution closer to this transformed value.

However, after thousands of MCMC iterations, the posterior distributions for both the stellar inclination and oblateness converged toward values numerically close to those reported by Cauley & Ahlers (2022), rather than the initially anticipated transformed configuration. This outcome was surprising and introduced some ambiguity, particularly because Cauley & Ahlers (2022) did not explicitly clarify their definition of the stellar inclination angle. Nevertheless, the consistent convergence of the MCMC chains toward these values indicated that, within the gravity-darkened photometric framework and the precision of the available TESS data, this configuration provided the most self-consistent explanation of the observed transit asymmetry. Despite the earlier concerns regarding coordinate conventions, the adopted values were therefore regarded as the most physically plausible solution supported by the present analysis.

With respect to the projected spin–orbit angle λ, our analysis yields a smaller value of $\lambda = 105.2 \pm 3.6^\circ$ compared to the value of approximately 113° reported by Cauley & Ahlers (2022) and Zhou et al. (2019). This difference arises directly from the MCMC-based gravity-darkening analysis performed in this work. As no Doppler tomography was used to independently constrain λ, the value obtained here should be interpreted with some caution and considered alongside the larger values reported in the literature. Notably, Bello-Arufe et al. (2022) also reported a smaller projected spin–orbit angle of $\lambda = 107.9^\circ$, which in some extent, is more consistent with the value derived in this study. Their result was obtained using Doppler tomography, a technique that provides a more direct constraint on the stellar surface velocity field during transit, explained in more details in the previous **Chapter 2.2**. Although their work primarily focused on characterising the chemical composition of the ultra-hot planet rather than investigating gravity darkening, their independent determination of a lower λ value lends some support to the result obtained here. Taken together, these findings suggest that a lower projected spin–orbit angle may be physically plausible for this system, potentially challenging the larger values reported in the literature. While the present analysis alone cannot definitively resolve this discrepancy, the consistency with Doppler tomography results provides motivation for further joint modelling combining both techniques.

A further point of disagreement concerned the gravity-darkening coefficient. Cauley & Ahlers (2022) adopted a relatively low value of $\beta = 0.20$, which could not be reproduced self-consistently within the rotational framework developed in this project. By applying the rotation-dependent prescription of Espinosa Lara & Rieutord (2011), and using the oblateness values inferred from the MCMC analysis, a higher gravity-darkening exponent of $\beta \approx 0.23$ was consistently obtained. As discussed in **Chapter 3.3.2** and illustrated in **Figure 12**, this value was physically expected for the inferred degree of stellar flattening. Furthermore, during the initial stages of this project, no combination of parameters was found that simultaneously reproduced the observed transit morphology while maintaining a gravity-darkening coefficient as low as $\beta = 0.20$. This discrepancy therefore most likely arose from differences in modelling assumptions rather than from limitations in the photometric data themselves, leaving scope for further debate regarding the optimal treatment of gravity darkening in extremely rapidly rotating stars.

Differences in the modelling approach also led to some disagreement in the determination of the polar and equatorial effective temperatures. Using the framework described in **Chapter 3.3**, the custom notebook developed for this study yielded values of $T_{\text{pole}} = 8307$ K and $T_{\text{equator}} = 8595$ K in the case when $i_\star \approx 55°$. Based on the parameters derived from the gravity-darkening analysis, a more self-consistent determination was obtained using the specialised exoplanet transit modelling code, ExoBush, which produced the values reported in **Table 3**: $T_{\text{pole}} = 8816$ K and $T_{\text{equator}} = 8242$ K. In addition to providing revised temperature estimates, the ExoBush analysis demonstrated strong consistency for several other key parameters, including the equatorial radius R_{eq}, the projected stellar rotational velocity $v\sin i_\star$, and the stellar oblateness f. These quantities were found to be very similar to those obtained in this study through the gravity-darkening exploration, lending confidence to the overall physical description of the stellar geometry. A key point of discussion is the discrepancy between these temperature estimates and those reported by Cauley & Ahlers, who derived $T_{\text{pole}} = 8670$ K and $T_{\text{equator}} = 7830$ K, albeit with relatively large uncertainties of approximately ± 300 K. The origin of this difference is likely linked to the modelling assumptions adopted in their analysis. As noted in previous sections, some aspects of their parameter derivation appear difficult to reconcile directly, which may contribute to the observed disagreement in the inferred temperature structure.

Finally, the adopted orbital inclination, stellar inclination, and projected spin–orbit angle were combined using **Equation 1** from Masuda (2015) to compute the true three-dimensional spin–orbit angle of the system. Solving this relation yielded a value of $\psi = 105.26°$. This result represented the primary scientific outcome of the present study and confirmed that HAT-P-70b occupies a strongly misaligned, retrograde orbit. The inferred geometry was consistent with formation via high-eccentricity migration pathways (*Winn et al., 2010; Albrecht et al., 2012; Rice et al., 2022*). Importantly, this result demonstrated that gravity-darkened transit modelling alone was capable of providing meaningful constraints on the full three-dimensional architecture of planetary systems orbiting rapidly rotating stars.

To conclude this section, a synthetic visualisation of the HAT-P-70 system was generated using the STARRY framework, adopting the final parameters reported in **Table 3**. This synthetic configuration illustrated the inferred stellar geometry, gravity-darkened surface-brightness distribution, and the planetary transit chord corresponding to the recovered orbit. The resulting visualisation is presented in **Figure 17** and provides an intuitive representation of the final system architecture inferred in this work.

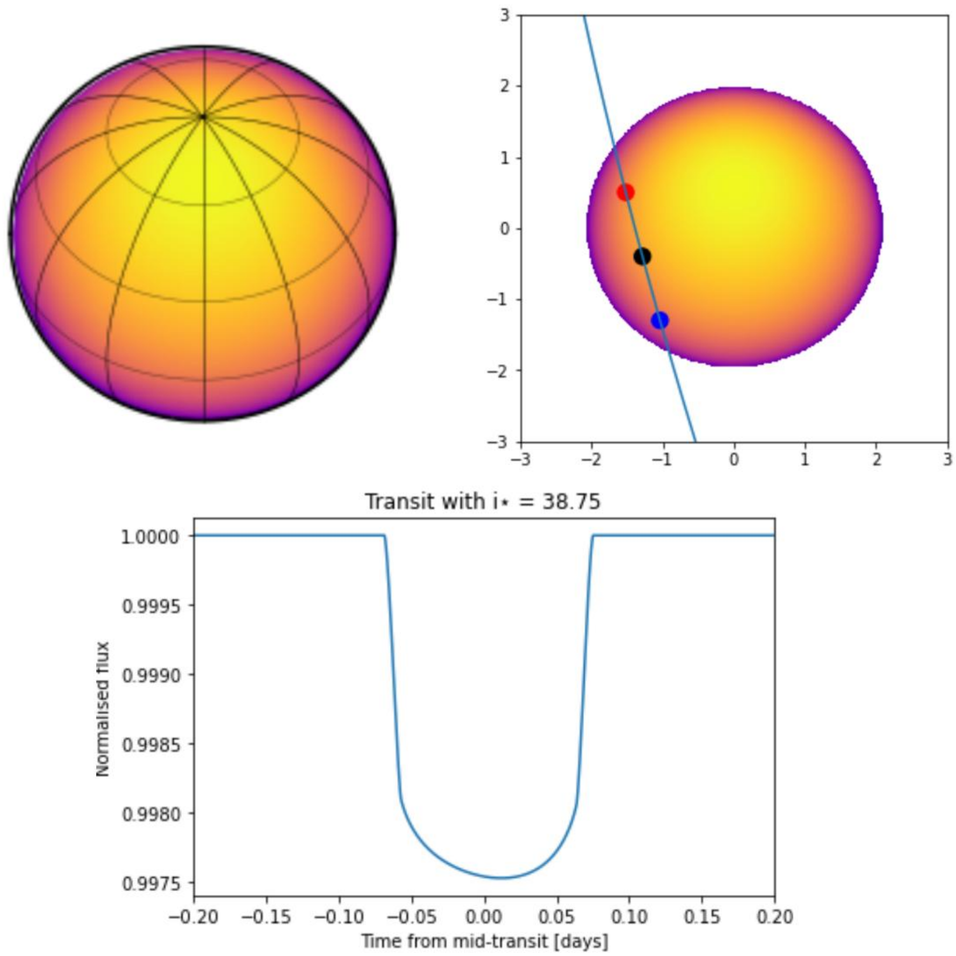

Figure 17. Final system geometry of the HAT-P-70 system derived from the gravity-darkened Markov Chain Monte Carlo analysis. The **top-left panel** shows a synthetic representation of the host star, illustrating the inferred stellar inclination, with the stellar rotation pole inclined at $i_{\star} = 38.75^{\circ}$ as obtained in this study. The **top-right panel** presents the projected transit geometry and transit chord across the stellar disk, with the planet moving from blue to red, indicating a strongly misaligned, retrograde orbit with $\lambda \approx 105^{\circ}$. The **bottom panel** shows the resulting synthetic gravity-darkened transit light curve produced by this configuration, which exhibits strong agreement with the observed transit morphology presented in **Figure 16**.

4.3 Comparative Case Studies and Target Selection

In addition to HAT-P-70, two further systems—NGTS-33 and MASCARA-1—were investigated at the initial stage of this project as potential candidates for gravity-darkened transit analysis. Both systems host planets orbiting hot, rapidly rotating stars and exhibit transit morphologies that deviate from the expectations of simple limb-darkened models. These deviations are more pronounced in MASCARA-1 than in NGTS-33. This section presents a comparative discussion of these systems, evaluates their suitability for detailed gravity-darkening studies, and explains the rationale for selecting HAT-P-70 as the primary and sole target of this work.

4.3.1 NGTS-33

NGTS-33 is a recently discovered hot-Jupiter system hosting the super-Jupiter NGTS-33b. The host star is a young, rapidly rotating A9V-type star with an effective temperature of T_{eff} = 7437 ± 72 K and a stellar rotation period of approximately 0.67 days, implying an extremely high projected rotational velocity of $v\sin i_{\star}$ ~110km/s (*Alves et al., 2025*). The planet has an orbital period of P = 2.83 days and a semi-major axis of a = 0.048 ± 0.002 au (*Alves et al., 2025*), resulting in frequent transits and strong star–planet interactions, which made the system particularly attractive for gravity-darkening investigations.

NGTS-33 b has a radius of R_{p} = 1.64 ± 0.07 R_{J} and a mass of 3.63 ± 0.27 M_{J}, placing it firmly in the super-Jupiter regime (*Alves et al., 2025*). Its large planetary radius produces a deep transit, which is advantageous for detailed transit-shape analyses compared to smaller planets whose signals are more susceptible to photometric noise. Combined with the host star's rapid rotation and high temperature, these properties made NGTS-33 a promising candidate for detecting gravity-darkening signatures in transit photometry.

TESS observations of NGTS-33 were available from two consecutive observing sectors, Sectors 87 and 88, obtained in 2024 and 2025, respectively. An initial inspection of the TESS processed light curves revealed clear transit events occurring approximately every three days. However, the individual transits appeared unusual: the transit depths varied systematically from event to event, producing a wave-like modulation in the in-transit flux, as shown in **Figure 18**. Before phase-folding, some transits appeared significantly deeper than others, suggesting time-dependent variability rather than a single, stable transit profile.

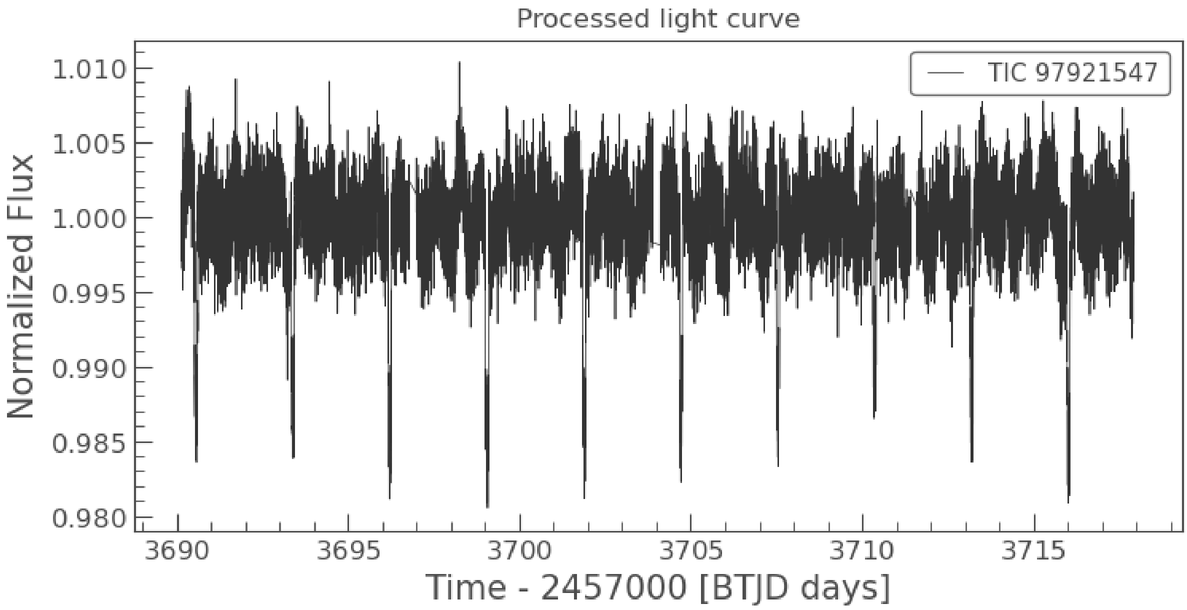

Figure 18. Processed TESS light curve of NGTS-33 using data from Sector 88 only. Clear transit events were observed approximately every 3 days; however, the individual transits exhibited variable depths. Some transits appeared deeper than others, and the overall in-transit flux displayed a wave-like modulation.

Such behaviour could plausibly arise from several mechanisms, including stellar activity associated with the rapid rotation of the host star, dynamical perturbations from an additional unseen companion, or residual instrumental systematics in the TESS photometry. After applying the same data-cleaning procedure as explained in **Chapter 3.1.2**—masking outliers, detrending, and phase-folding—the resulting phase-folded transit did not exhibit a strong or persistent asymmetry. Despite this, the combination of the star's extreme rotational velocity, high effective temperature, and the planet's short orbital period suggested that NGTS-33 remained an interesting target for gravity-darkening analysis.

Standard limb-darkened transit fitting was first performed using literature parameters from Alves et al. (2025). This approach yielded a relatively poor fit to the phase-folded TESS data, with a $\chi^2_{\text{red}} = 2.06$. Allowing key geometric parameters—specifically the planet-to-star radius ratio, the scaled semi-major axis, and the impact parameter—to vary freely resulted in a substantial improvement, reducing the χ^2_{red} to a value close to unity. This result indicated that, within the precision of the available TESS photometry, the observed transit shape of NGTS-33 b could be adequately reproduced without invoking gravity darkening.

Nevertheless, the high level of photometric noise and the wave-like variability observed in the individual transits were not fully accounted for by the phase-folding process. These effects likely diluted or masked any subtle gravity-darkening signature that may have been present. With additional observing time, higher signal-to-noise photometry, or dedicated spectroscopic follow-up, NGTS-33 could become an exceptionally valuable system for probing stellar obliquity through gravity darkening or Doppler tomography.

In summary, although NGTS-33 remained a promising candidate for future spectroscopic and Doppler-tomographic follow-up, the absence of a robust and persistent gravity-darkening signature in the available photometry rendered it less suitable for a dedicated gravity-darkened transit analysis at the present time. The observed variability introduced significant degeneracies that were difficult to resolve using photometry alone, motivating the prioritisation of other targets in this study.

4.3.2 MASCARA-1

MASCARA-1 represents the opposite extreme to NGTS-33 in terms of prior characterisation, as the system has been studied extensively in the literature. It is known to host a hot Jupiter on a strongly misaligned orbit around a rapidly rotating A-type star, and multiple studies have demonstrated that gravity darkening plays a dominant role in shaping its transit light curve (*Talens et al. 2017; Hooton et al. 2022*). The host star has an effective temperature of approximately $T_{\rm eff} \sim 7490$ K (*Talens et al. 2017*) and a very high projected rotational velocity $v\sin i_\star$ $\sim$109 km/s (*Talens et al. 2017*), making it highly oblate and therefore an ideal laboratory for gravity-darkening effects. The planet MASCARA-1 b has an orbital period of P ≈ 2.15 days and a large radius $R_{\rm p} \sim 1.5\ R_{\rm J}$, resulting in deep, high signal-to-noise transits that are well suited for detailed transit-shape analyses (*Talens et al. 2017*).

The analysis of two high-quality TESS sectors—Sector 55 (2022) and Sector 82 (2024)—revealed clear and regularly spaced transit events occurring approximately every two days. After cleaning the data and phase-folding the light curves, a pronounced and repeatable asymmetry was observed. This asymmetric morphology is a signature of gravity darkening in combination with a misaligned planetary orbit, and it was already evident at the level of the phase-folded photometry.

When the transits were fitted using a standard limb-darkened transit model with literature parameters held fixed, the resulting reduced $\chi^2_{\rm red}$ exceeded 6, clearly indicating that the model failed to reproduce the observed transit morphology. Even after allowing key geometric parameters to vary freely, the fit quality improved only marginally, with $\chi^2_{\rm red}$ remaining above 2 and structured residuals persisting throughout the transit. These residuals were strongly indicative of missing physics in the model and provided compelling evidence for the presence of gravity darkening and significant spin–orbit misalignment, in agreement with previous studies (*Talens et al. 2017; Hooton et al. 2022*).

From a purely diagnostic perspective, MASCARA-1 represents an excellent and unambiguous example of a gravity-darkened transit. However, the system has already been analysed in detail in the literature using advanced modelling techniques, including gravity-darkened transit fitting and Doppler tomography. As a result, any further analysis performed here would primarily serve to reproduce or validate existing results rather than provide new constraints or methodological insight.

Given the objectives of this project—to independently assess gravity-darkened photometry and explore its constraining power in systems that have not yet been exhaustively studied—this substantially reduced the scientific novelty of selecting MASCARA-1 as the primary case study. Consequently, despite its clear and striking gravity-darkening signature, MASCARA-1 was not chosen for detailed follow-up analysis in this work.

4.4 Implications for Formation and Migration History

The strongly misaligned and retrograde spin–orbit geometry inferred for the HAT-P-70 system provides a powerful constraint on its formation and migration history. Gravity-darkened transit modelling presented in **Chapter 4** yields a large three-dimensional spin–orbit angle of ψ = 105.26°, indicating that the planetary orbital plane is significantly tilted relative to the stellar rotation axis. Because large spin–orbit angles are difficult to modify once a planet has migrated to a close-in orbit, the observed geometry is most likely indicative of the dominant migration mechanism rather than the result of subsequent tidal evolution.

Formation pathways that preserve primordial alignment, such as disk-driven migration or in-situ formation, are not naturally expected to produce such an extreme configuration. Disk migration generally maintains spin–orbit alignment, as gravitational torques between the planet and the protoplanetary disk act within a single, well-defined plane, unless the disk itself is substantially tilted by external perturbations (*Lin et al., 1996; Batygin, 2012*). Similarly, in-situ formation models predict that hot Jupiters form and remain close to the disk midplane, with large spin–orbit misalignments arising only under specific circumstances, such as primordial star–disk misalignment (*Batygin et al., 2016*). While such effects may contribute to moderate obliquities, they do not readily generate strongly retrograde configurations, rendering them unlikely explanations for the HAT-P-70 system.

High-eccentricity migration provides a more natural explanation for the observed geometry. Dynamical mechanisms including planet–planet scattering (*Chatterjee et al., 2008; Nagasawa et al., 2008*), Kozai–Lidov oscillations driven by distant companions (*Wu & Murray, 2003; Fabrycky & Tremaine, 2007*), and secular chaos (*Wu & Lithwick, 2011*) are capable of exciting both large orbital eccentricities and inclinations through chaotic exchanges of angular momentum that do not preserve alignment with the stellar spin axis. Following these interactions, tidal dissipation at small periastron distances circularises the orbit while largely preserving the orbital inclination, resulting in close-in giant planets on misaligned or retrograde orbits.

The long-term survival of such misalignments depends sensitively on stellar properties. Tidal realignment of the stellar spin axis is efficient primarily in cool stars with substantial convective envelopes, where strong magnetic braking enhances tidal dissipation (*Dawson, 2014*). In contrast, hot stars above the Kraft break experience weak magnetic braking and inefficient tidal dissipation, leading to realignment timescales that exceed typical system ages. With an effective temperature of approximately 8450 K, HAT-P-70 lies well above the Kraft break, making the persistence of a large ψ fully consistent with theoretical expectations.

Population-level studies further support this interpretation. Rice et al. (2022) demonstrated that hot Jupiters on eccentric orbits exhibit a broad distribution of stellar obliquities that is largely independent of host star temperature, consistent with high-eccentricity migration. While circular hot Jupiters orbiting cool stars frequently appear aligned due to efficient tidal damping, systems hosted by hot stars retain their primordial or dynamically induced misalignments. Within this framework, the observed geometry of HAT-P-70 is best interpreted as a preserved post-migration configuration rather than the product of significant tidal reprocessing.

Overall, the extreme spin–orbit misalignment recovered for the HAT-P-70 system strongly disfavours disk migration or simple in-situ formation as the dominant evolutionary pathway. Instead, the observations are most consistent with high-eccentricity migration driven by strong dynamical interactions, followed by tidal circularisation of the orbit. HAT-P-70 therefore represents a compelling example of a hot-Jupiter system whose three-dimensional architecture retains the imprint of its dynamical migration history.

5 Conclusion

This project has explored the extent realised gravity-darkened transit photometry can constrain the three-dimensional spin–orbit geometry of a hot-Jupiter system orbiting a rapidly rotating star, using HAT-P-70 as a case study. The analysis shows that the available TESS photometry contains a measurable, but relatively subtle, asymmetric transit signal that is broadly consistent with expectations from gravity darkening, while remaining less pronounced than in more extreme systems such as KELT-9 and MASCARA-1.

The modelling demonstrates that gravity-darkened transit fits can reproduce the overall transit morphology of HAT-P-70 and provide useful constraints on the system geometry, particularly in a regime where spectroscopic techniques are inherently limited. However, the relatively modest amplitude of the asymmetry leads to significant degeneracies between stellar inclination, gravity-darkening strength, and orbital geometry. As a result, multiple physically plausible configurations remain compatible with the data, and the inferred parameters must be interpreted with appropriate caution.

Residual structure within the transit, most notably near mid-transit, suggests that the system is not yet fully described by the adopted models. These discrepancies may reflect limitations in the current treatment of stellar oblateness, gravity darkening, or surface-brightness structure, or may simply arise from the finite precision and temporal coverage of the available photometry. Rather than indicating a failure of the modelling approach, these features emphasise the complexity of the system and highlight the need for improved data and refined modelling assumptions.

In this context, HAT-P-70 should be regarded as an informative and challenging target rather than an archetypal example of gravity-darkened transits. Its combination of rapid stellar rotation and moderate gravity-darkening signature places it in an intermediate regime between weakly rotating systems and extreme cases such as KELT-9. This makes it particularly valuable for assessing the sensitivity and limitations of gravity-darkened transit modelling when the signal is present but not dominant.

The forthcoming release of additional TESS observations from Sector 98 offers a clear pathway for future progress. Increased photometric precision and a larger number of transits will allow the persistence and origin of the observed asymmetries to be tested more rigorously, potentially reducing current degeneracies and clarifying the nature of the unexplained transit structure. Continued study of HAT-P-70 with improved data therefore remains well motivated and may provide further insight into the applicability of gravity-darkened transit modelling across a broader range of stellar and planetary systems.

6 Appendices

6.1 Appendix A - MCMC Trace Plot Results for Configurations A, B, and C

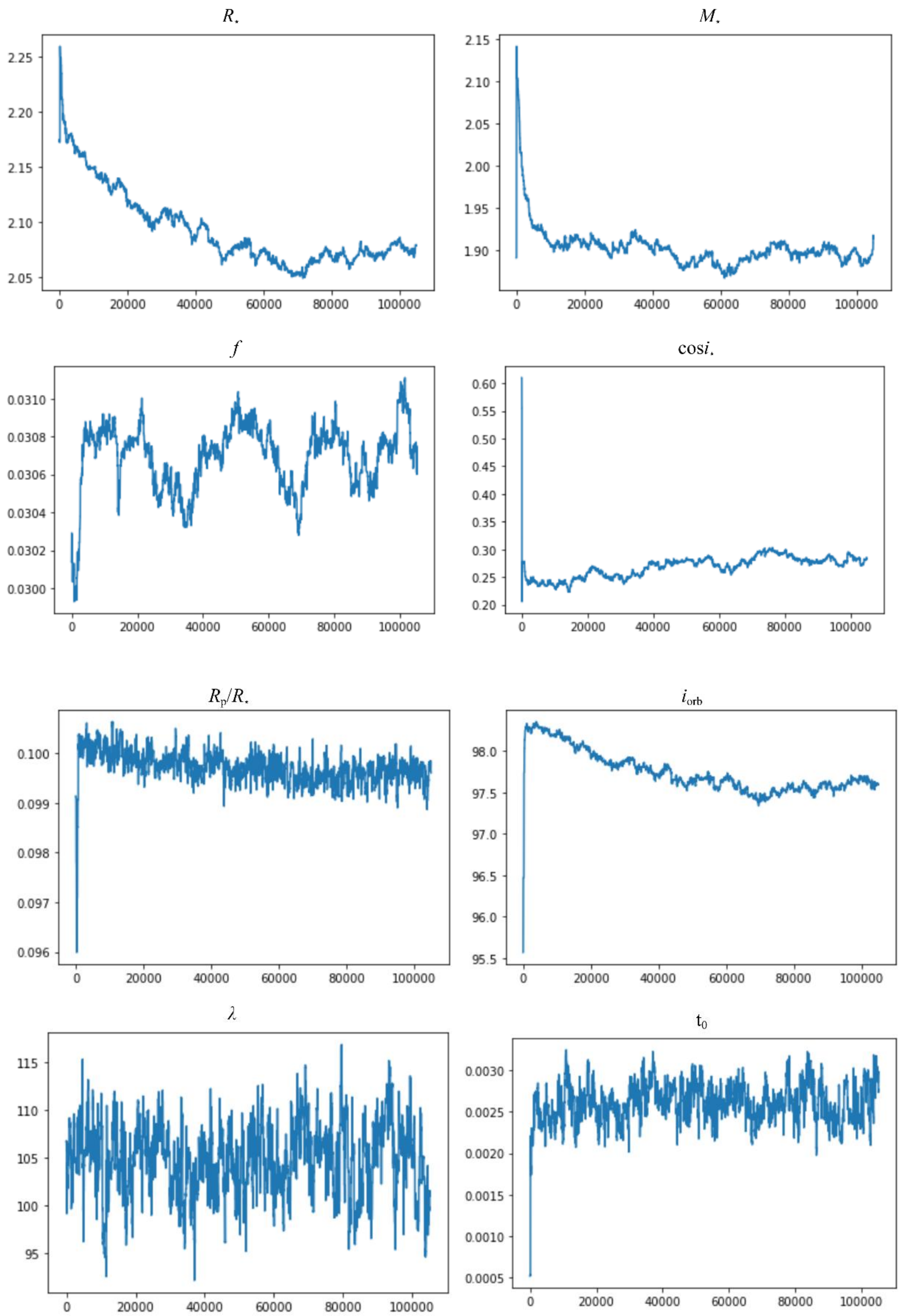

Figure 1A. Trace plots for ***Configuration A*** obtained after approximately 150,000 MCMC iterations using the adapted MCMC framework. The chains explore the posterior behaviour of the stellar and orbital parameters, including the stellar radius ($R_\star$), stellar mass ($M_\star$), stellar oblateness (f), cosine of the stellar inclination ($\cos i_\star$), planet-to-star radius ratio ($R_p/R_\star$), orbital inclination (i_{orb}), projected spin–orbit angle (λ), and mid-transit time (t_0). In each panel, the x-axis represents the MCMC iteration number, while the y-axis shows the sampled value of the corresponding model parameter.

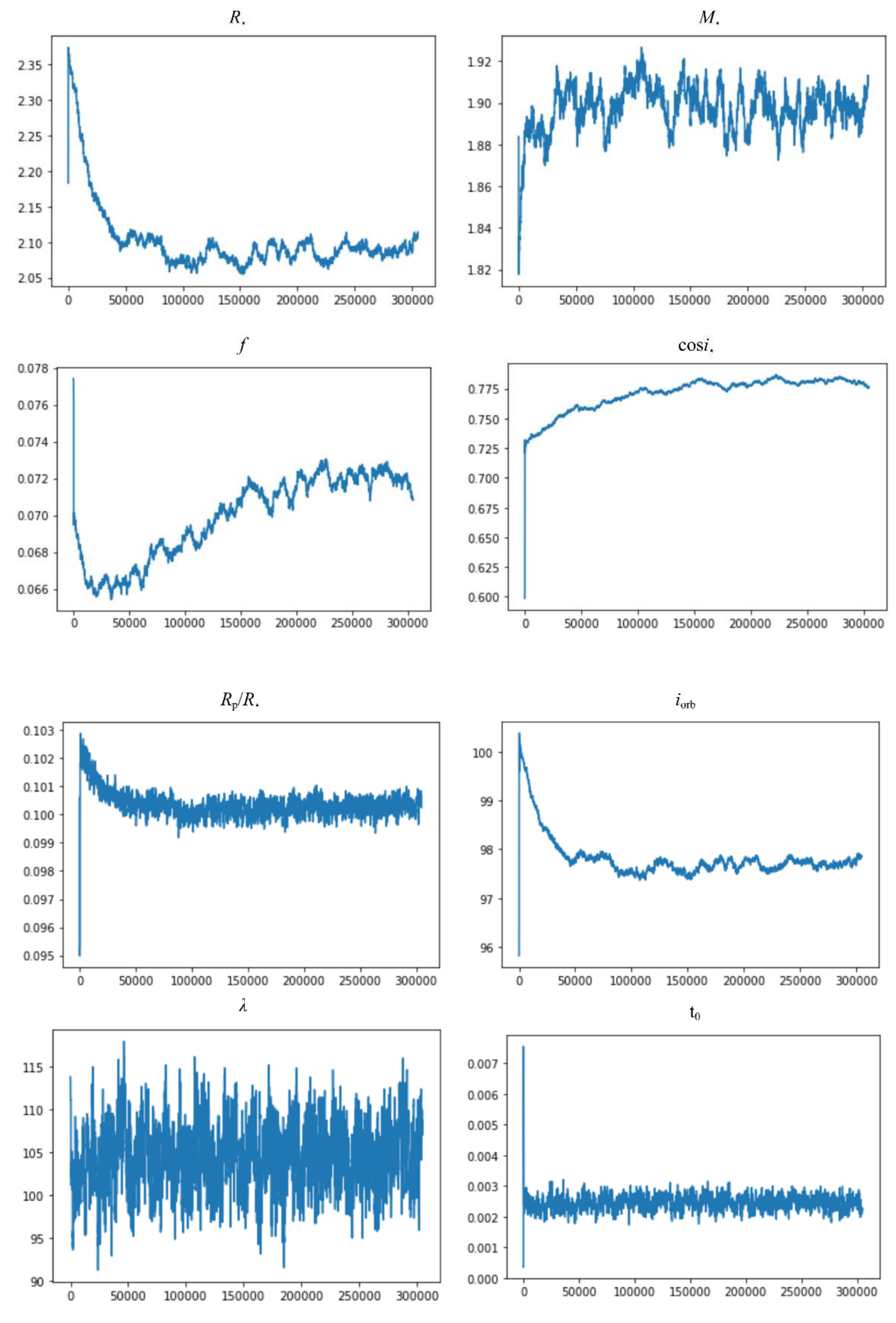

Figure 2A. Trace plots for ***Configuration B*** obtained after approximately 300,000 MCMC iterations using the adapted MCMC framework. The chains explore the posterior behaviour of the stellar and orbital parameters, including the stellar radius ($R_\star$), stellar mass ($M_\star$), stellar oblateness (f), cosine of the stellar inclination ($\cos i_\star$), planet-to-star radius ratio ($R_p/R_\star$), orbital inclination (i_{orb}), projected spin–orbit angle (λ), and mid-transit time (t_0). In each panel, the x-axis represents the MCMC iteration number, while the y-axis shows the sampled value of the corresponding model parameter.

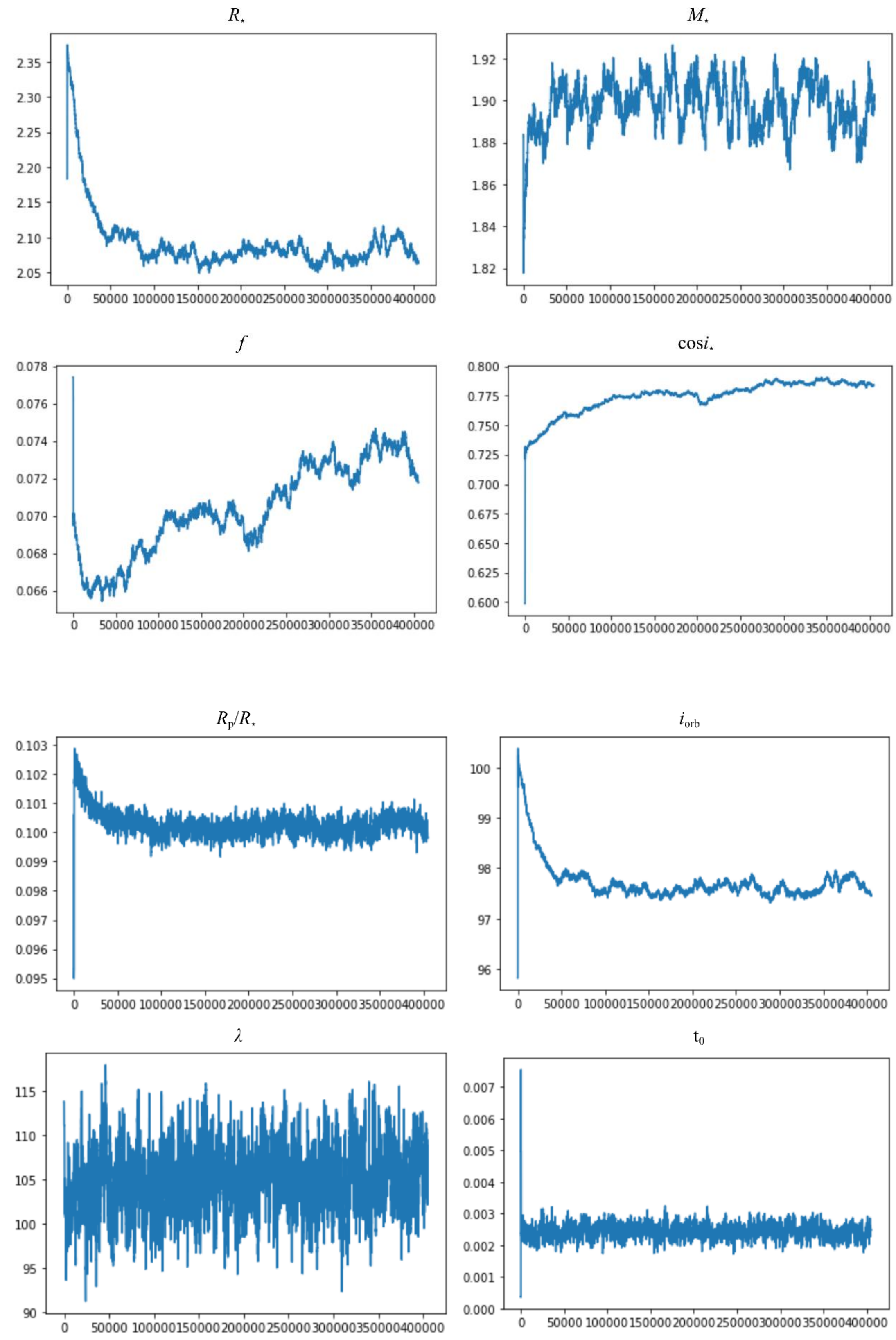

Figure 3A. Trace plots for ***Configuration C*** obtained after approximately 400,000 MCMC iterations using the adapted MCMC framework. The chains explore the posterior behaviour of the stellar and orbital parameters, including the stellar radius ($R_\star$), stellar mass ($M_\star$), stellar oblateness (f), cosine of the stellar inclination ($\cos i_\star$), planet-to-star radius ratio ($R_p/R_\star$), orbital inclination (i_{orb}), projected spin–orbit angle (λ), and mid-transit time (t_0). In each panel, the x-axis represents the MCMC iteration number, while the y-axis shows the sampled value of the corresponding model parameter.

6.2 Appendix B – MCMC Corner Plot Results for Configuration A, B and C

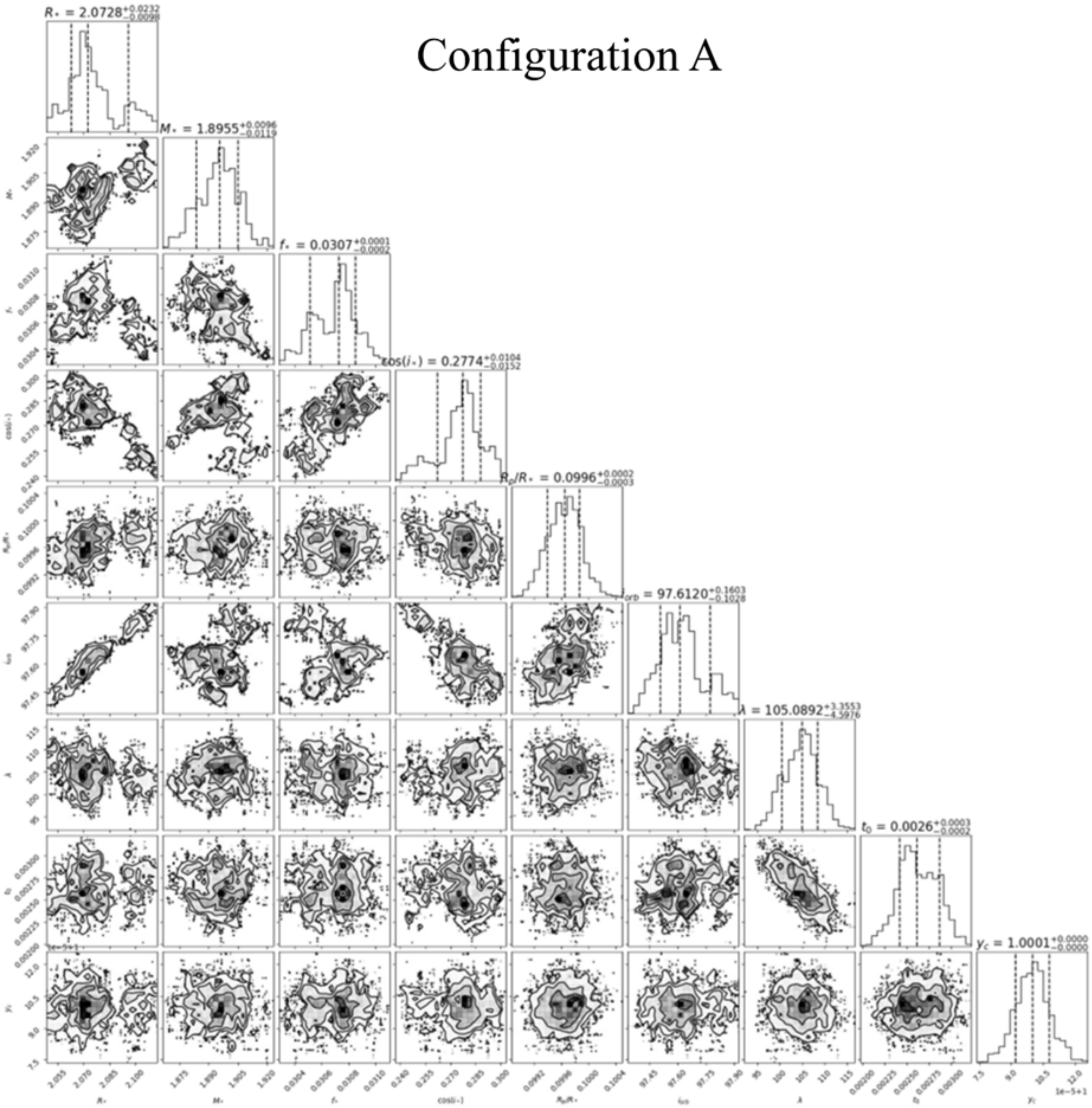

Figure B1. Corner plot showing the posterior distributions and parameter correlations for ***Configuration A*** obtained from the MCMC analysis. The one-dimensional histograms along the diagonal represent the marginalised posterior distributions for each parameter, with dashed lines indicating the median and credible intervals. The two-dimensional panels show the joint posterior distributions and correlations between parameters, with contours corresponding to increasing confidence levels. The explored parameters include the stellar radius ($R_\star$), stellar mass ($M_\star$), stellar oblateness (f), cosine of the stellar inclination ($\cos i_\star$), planet-to-star radius ratio ($R_p/R_\star$), orbital inclination (i_{orb}), projected spin–orbit angle (λ), and mid-transit time (t_0).

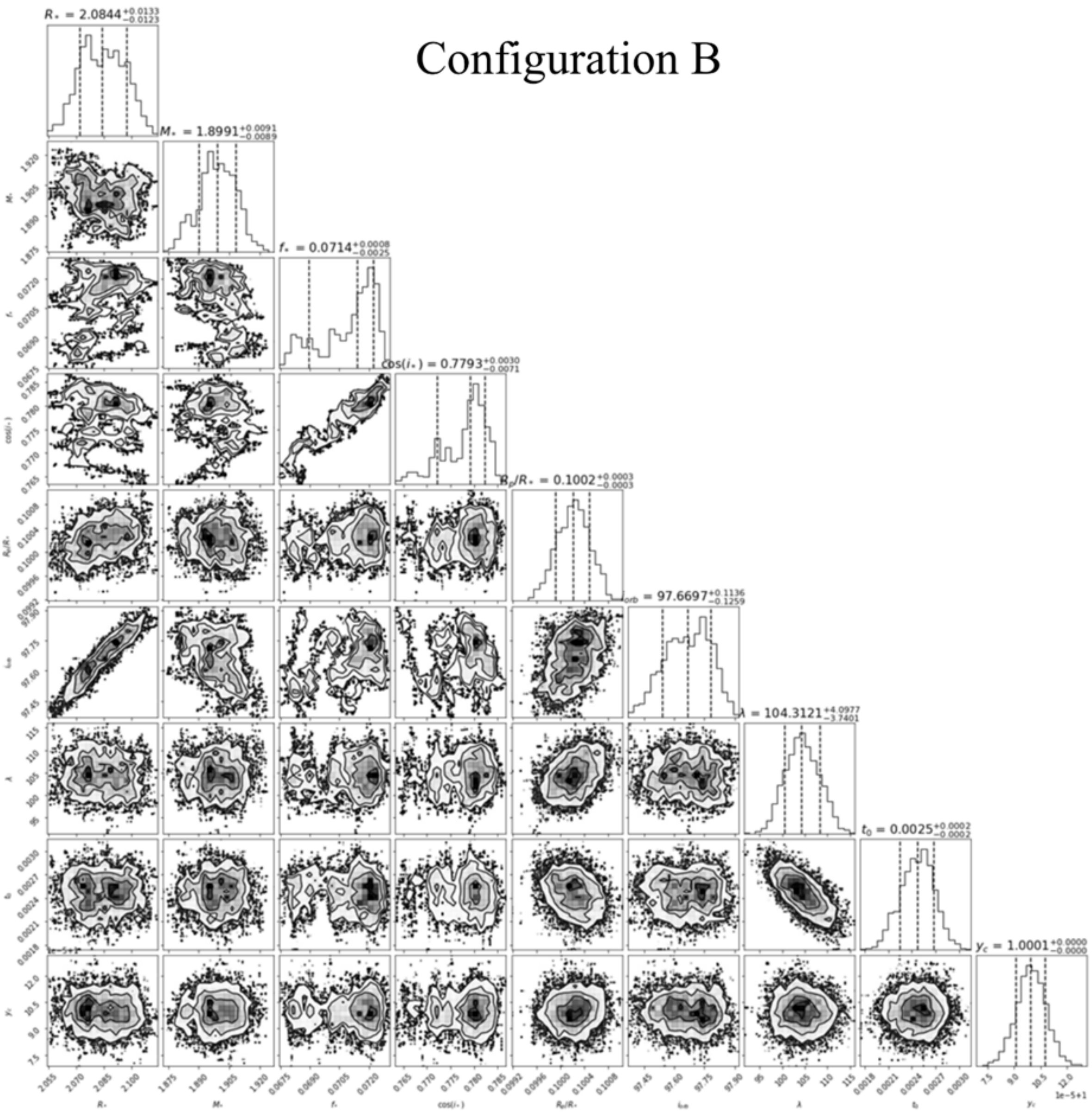

Figure B2. Corner plot showing the posterior distributions and parameter correlations for ***Configuration B*** obtained from the MCMC analysis. The one-dimensional histograms along the diagonal represent the marginalised posterior distributions for each parameter, with dashed lines indicating the median and credible intervals. The two-dimensional panels show the joint posterior distributions and correlations between parameters, with contours corresponding to increasing confidence levels. The explored parameters include the stellar radius ($R_\star$), stellar mass ($M_\star$), stellar oblateness (f), cosine of the stellar inclination ($\cos i_\star$), planet-to-star radius ratio ($R_p/R_\star$), orbital inclination (i_{orb}), projected spin–orbit angle (λ), and mid-transit time (t_0).

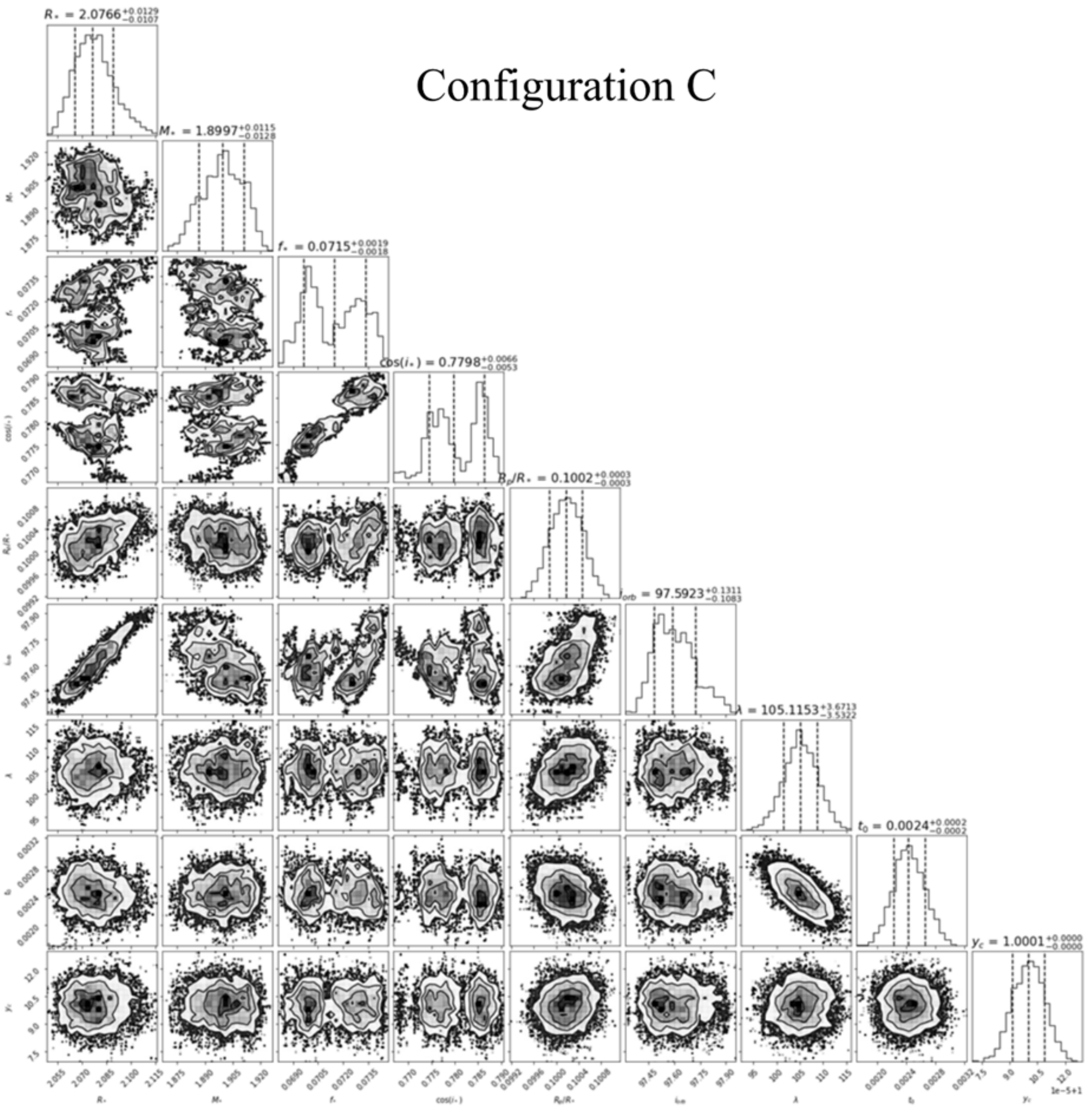

Figure B3. Corner plot showing the posterior distributions and parameter correlations for ***Configuration C*** obtained from the MCMC analysis. The one-dimensional histograms along the diagonal represent the marginalised posterior distributions for each parameter, with dashed lines indicating the median and credible intervals. The two-dimensional panels show the joint posterior distributions and correlations between parameters, with contours corresponding to increasing confidence levels. The explored parameters include the stellar radius ($R_\star$), stellar mass ($M_\star$), stellar oblateness (f), cosine of the stellar inclination ($\cos i_\star$), planet-to-star radius ratio ($R_p/R_\star$), orbital inclination (i_{orb}), projected spin–orbit angle (λ), and mid-transit time (t_0).

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